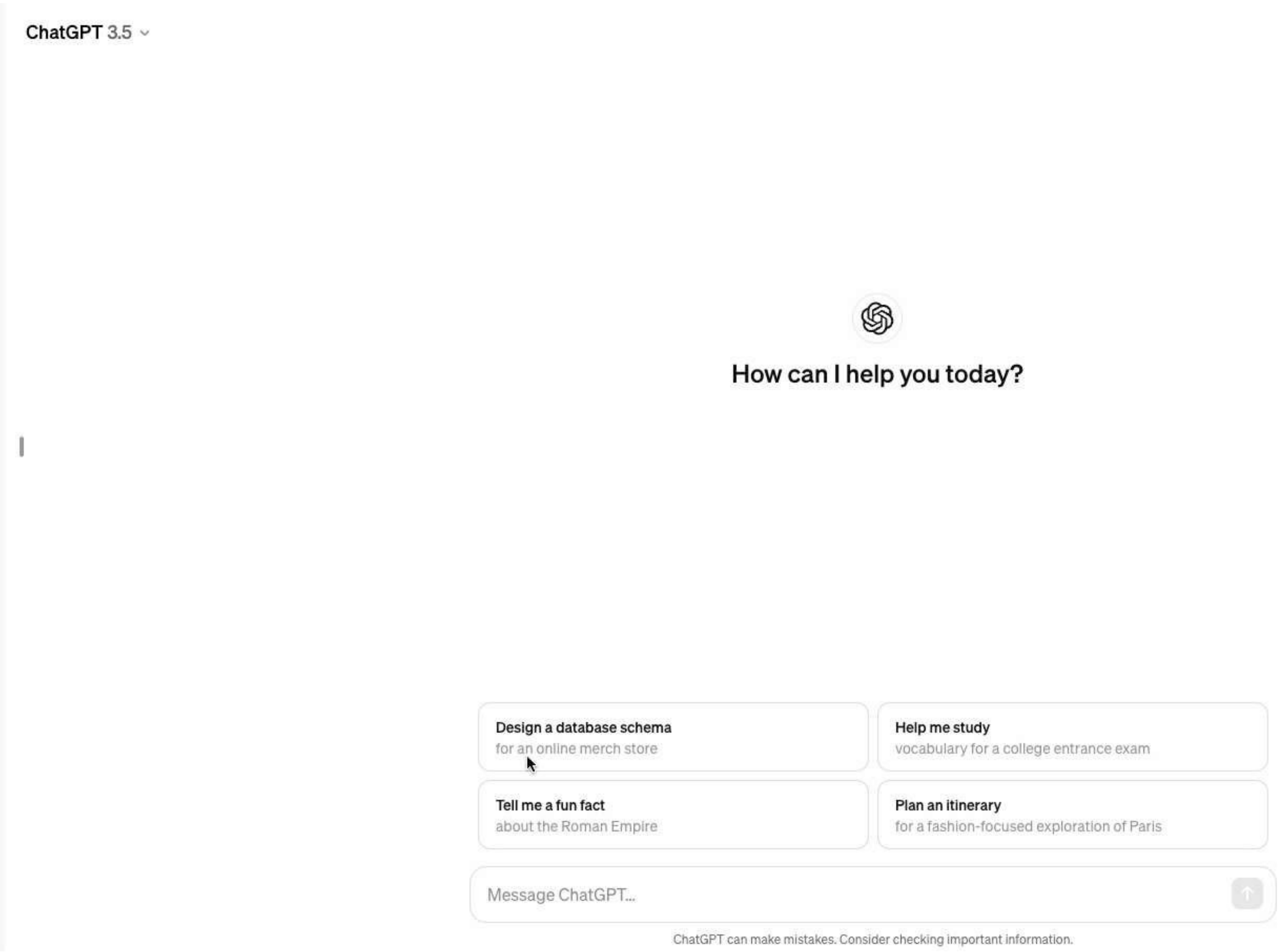
Efficient model Inference

ChatGPT


ChatGPT 3.5
How can I help you today?
Design a database schema
for an online merch store
Help me study
vocabulary for a college entrance exam
Tell me a fun fact
about the Roman Empire
Plan an itinerary
for a fashion-focused exploration of Paris
Message ChatGPT...
ChatGPT can make mistakes. Consider checking important information.

TTFT

Time To First Token

Context decoding

Tokens Per Second

You
Tell me a random fun fact about the Roman Empire

ChatGPT
Sure! Did you know that the ancient Romans used urine as a cleaning agent? They believed that the ammonia in urine could help whiten and brighten their clothes. They even col ●

Generation step

TPS

Tokens Per Second

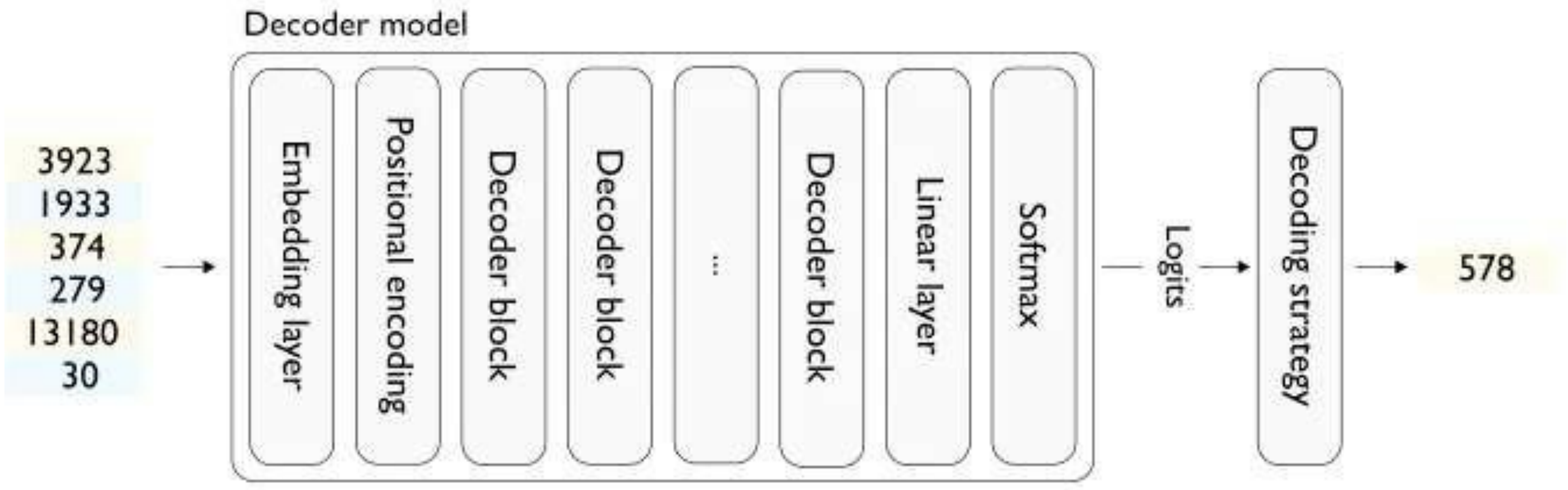
	TPS for single query	TPS for single instance
Assistant	40 TPS is OK (or is it?)	Economics
Offline	Do we care?	Most efficient way

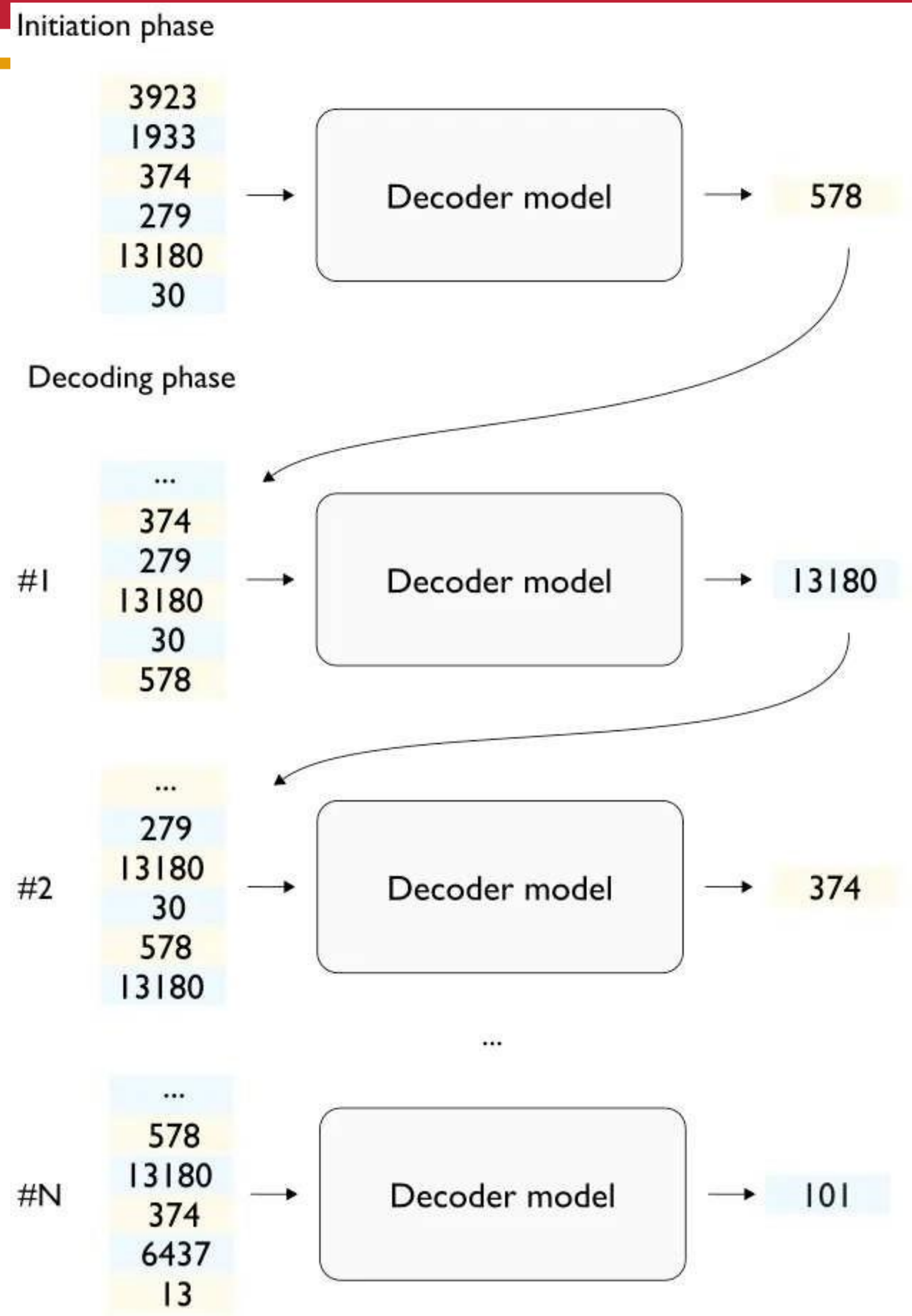
Inference

- Prompt processing
- Autoregressive steps
- Hardware utilization

- Serving queries
- Business cases

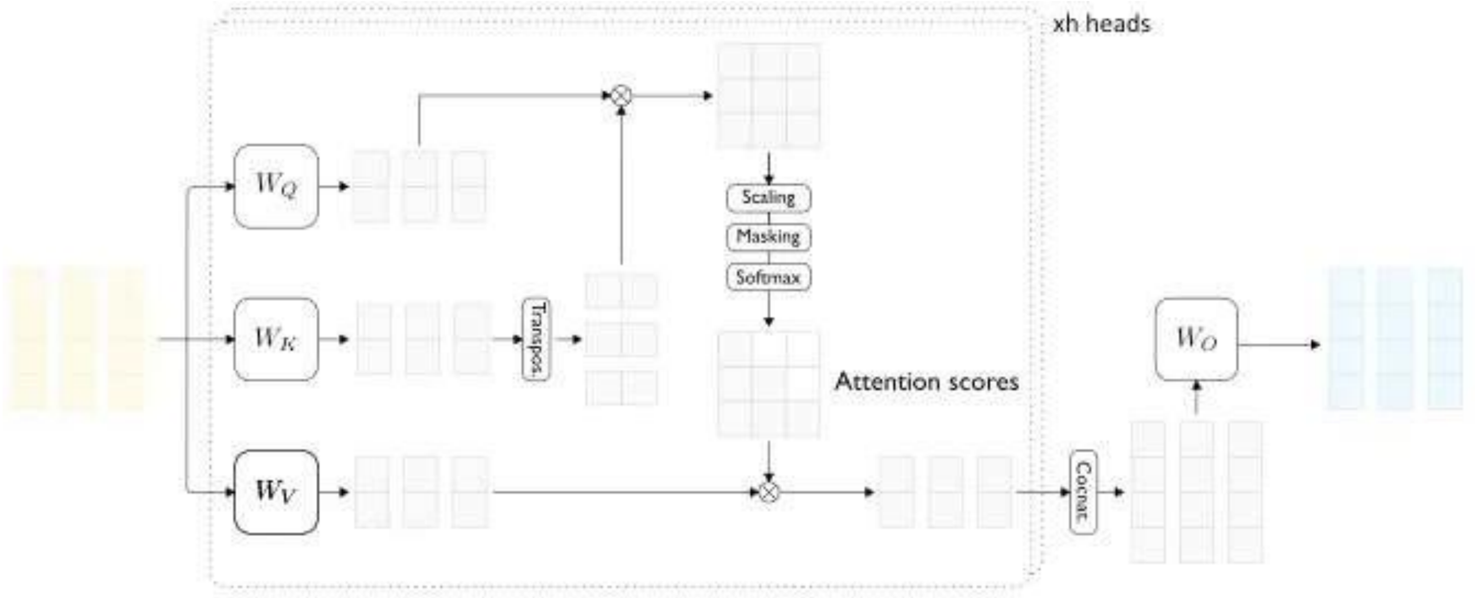
LLM response

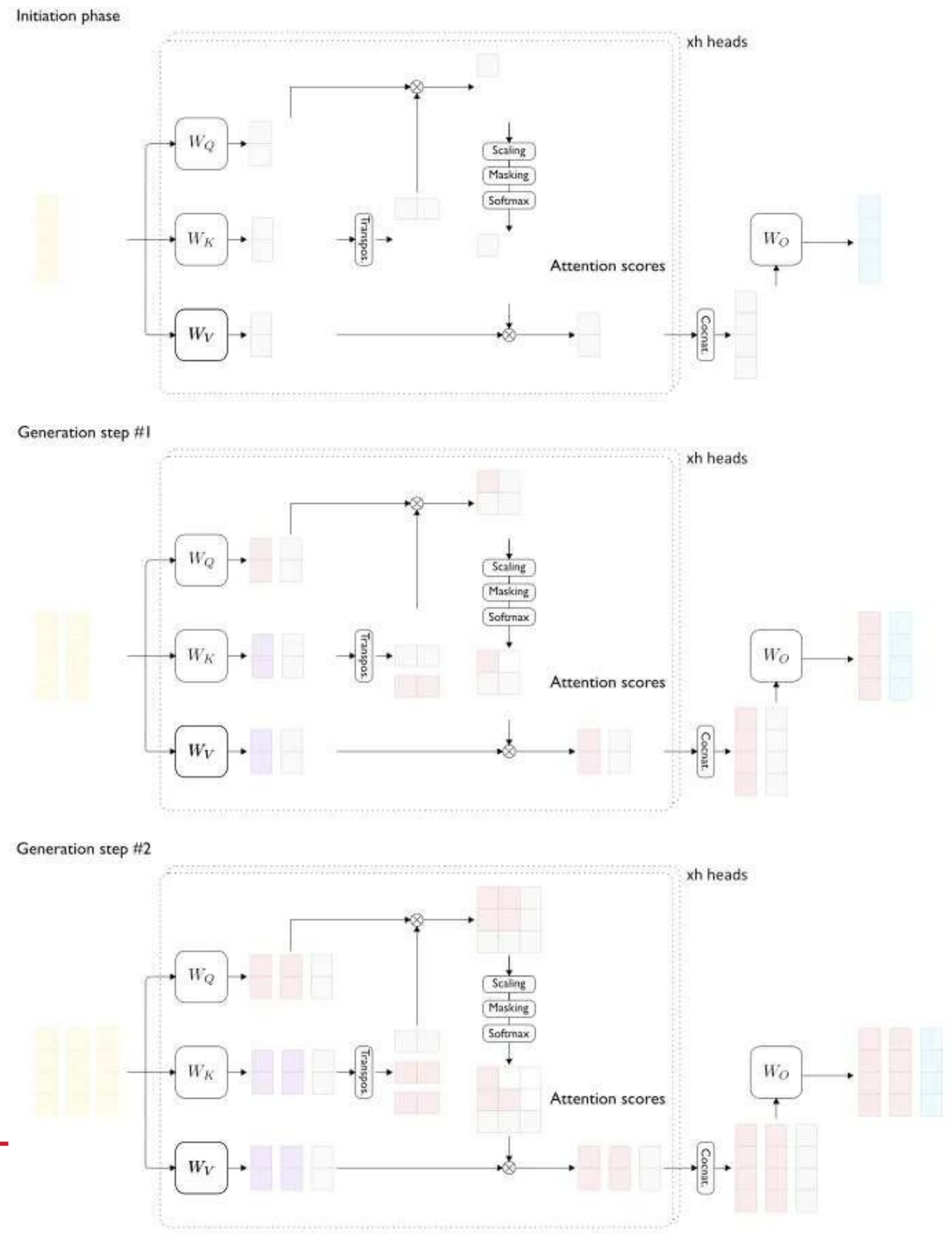




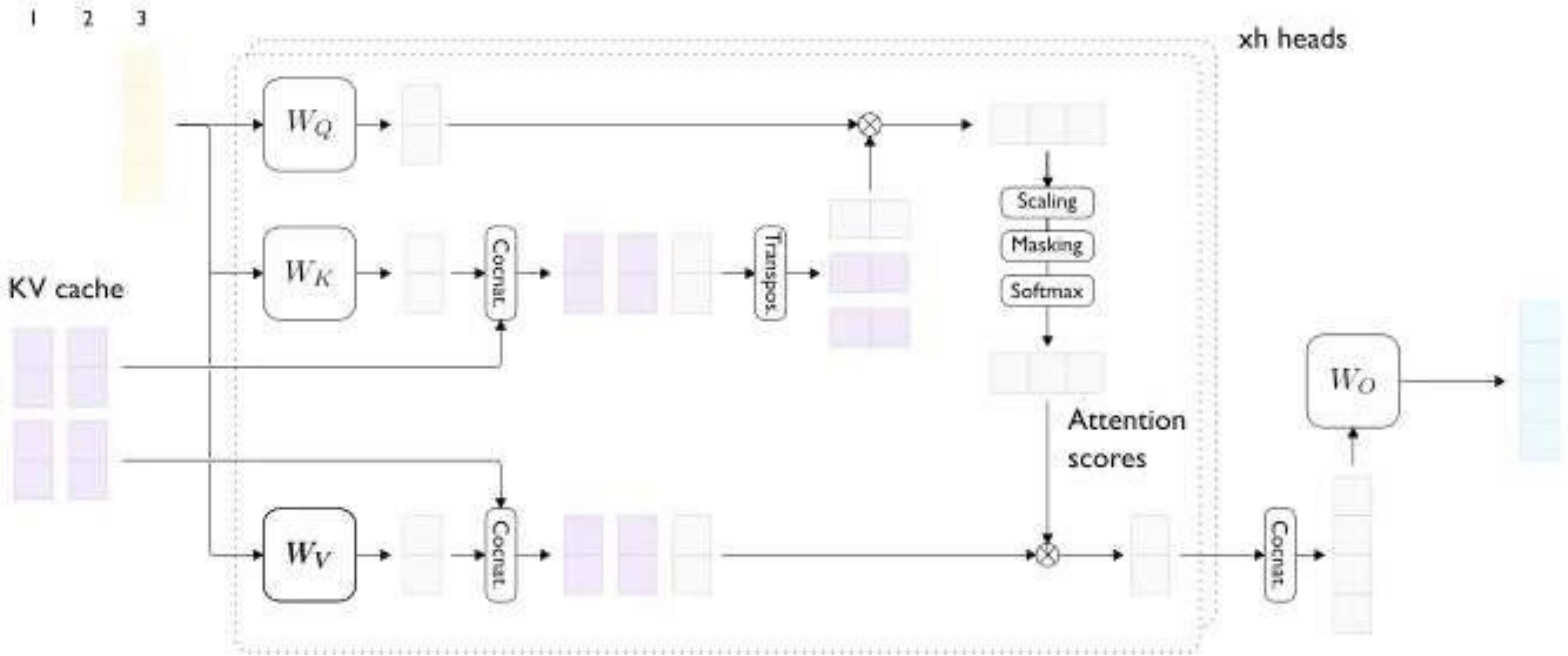
KV cache

Decoder block

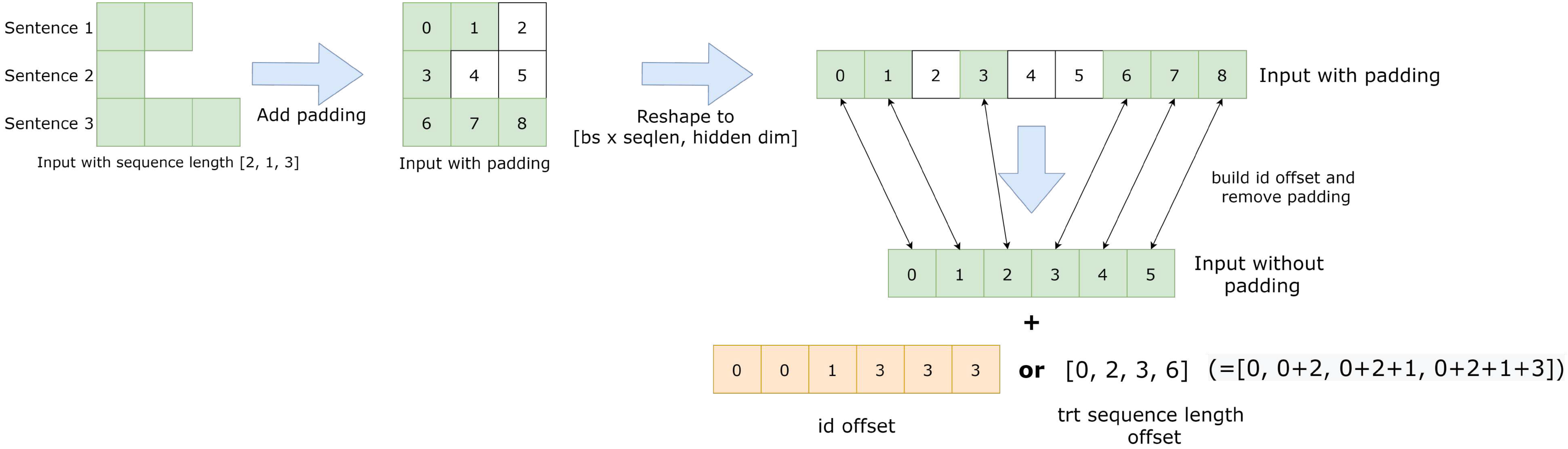




KV cache

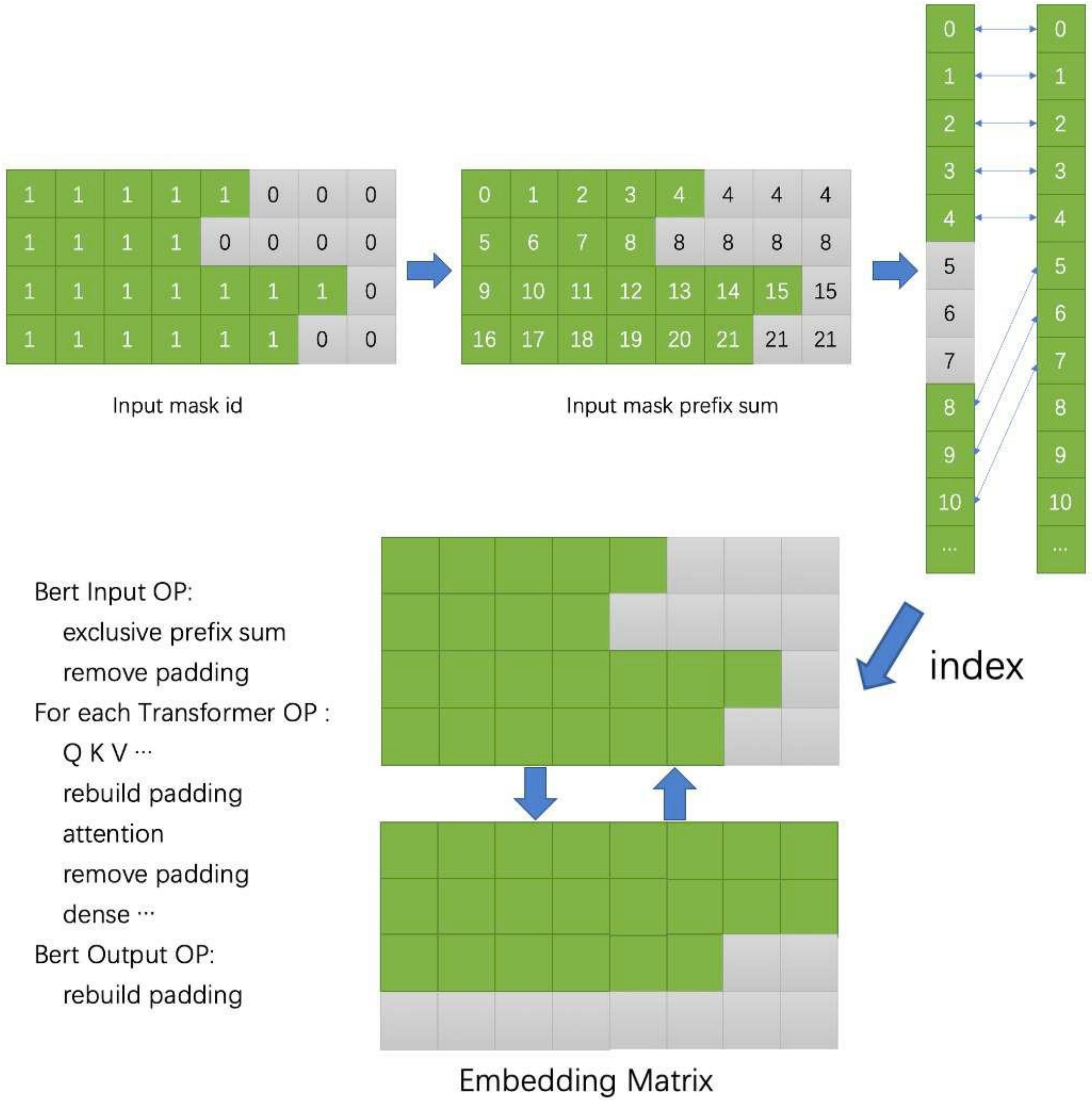

1
2
3
xh heads
W_Q
W_K
W_V
W_O
KV cache
Concat.
Transpos.
Scaling
Masking
Softmax
Attention scores
Concat.
Concat.

Batch



Source: https://github.com/bytedance/effective_transformer

Batch



Source: https://github.com/bytedance/effective_transformer

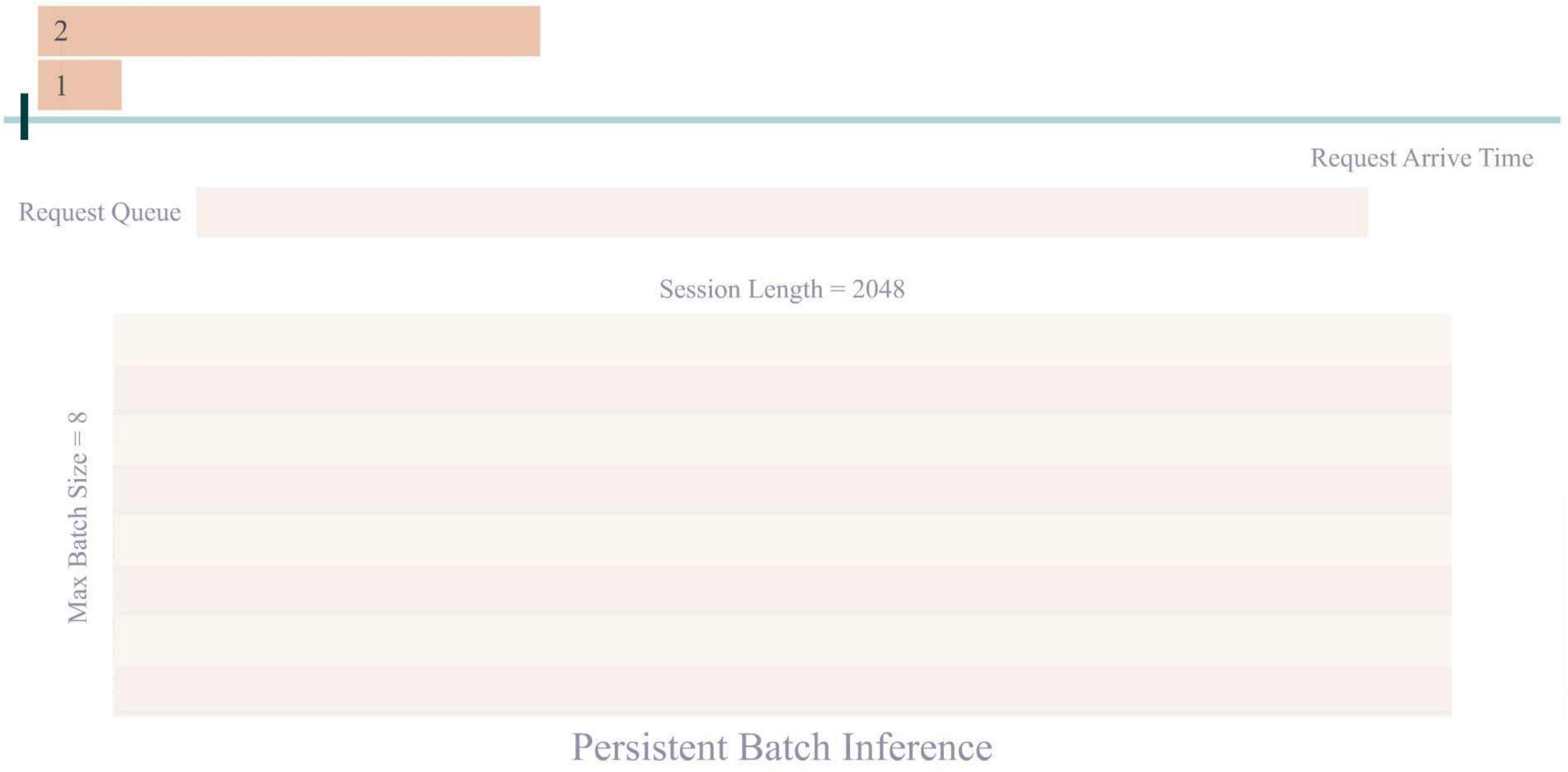
Batch

Tesla V100, float16, maximum sequence length=64, average sequence length≈40

batch_size	XLA (in ms)	Faster Transformer (in ms)	Speedup over XLA	Effective Transformer (in ms)	Speedup over XLA
100	28.31	20.27	1.40	16.03	1.77
200	54.47	40.08	1.36	30.15	1.81
300	80.53	59.11	1.36	41.27	1.95
400	106.5	78.38	1.36	54.12	1.97
500	132.35	98.03	1.37	65.92	2.01
1000	261.18	190.91	1.38	133.61	1.95

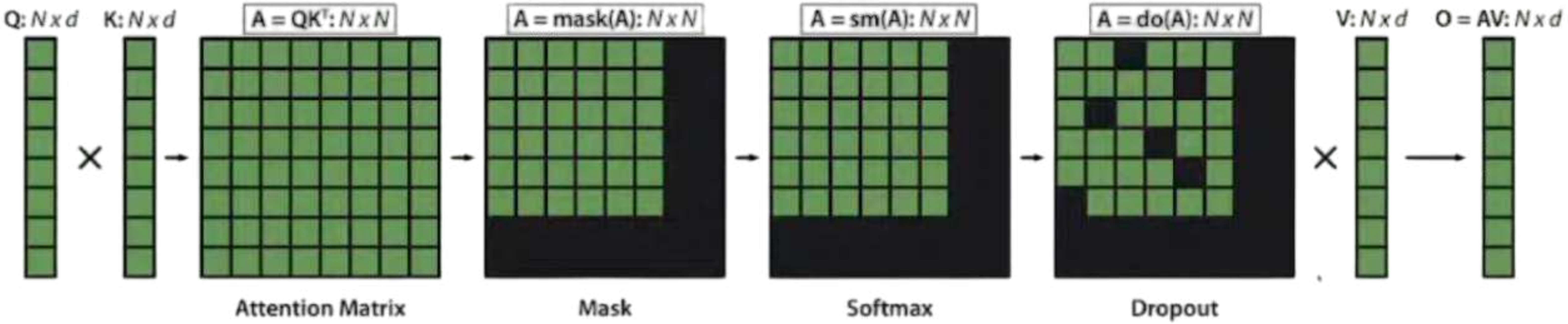
Source: https://github.com/bytedance/effective_transformer

Continuous batch



Source: https://github.com/InternLM/lmdeploy

Flash attention

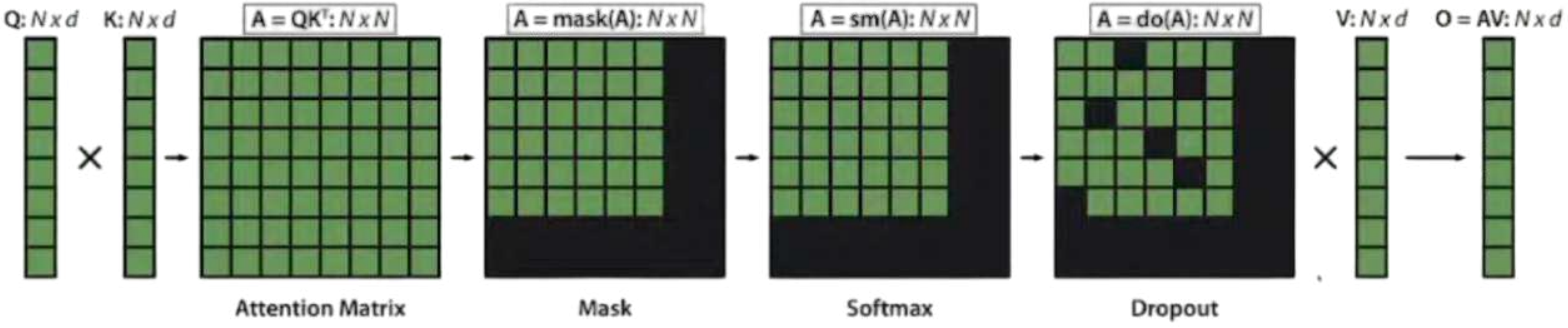


$$\mathbf{O} = \mathrm{Dropout}(\mathrm{Softmax}(\mathrm{Mask}(\mathbf{QK}^T)))\mathbf{V}$$

Naive implementation requires repeated R/W from slow GPU HBM. Hard to scale to long sequences

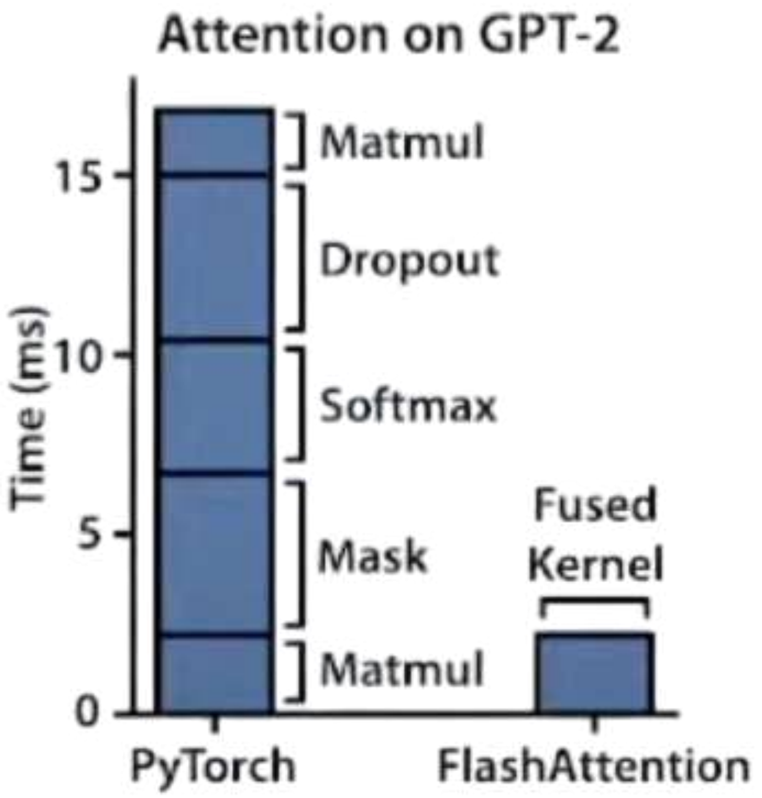
Source: https://arxiv.org/abs/2205.14135

Flash attention



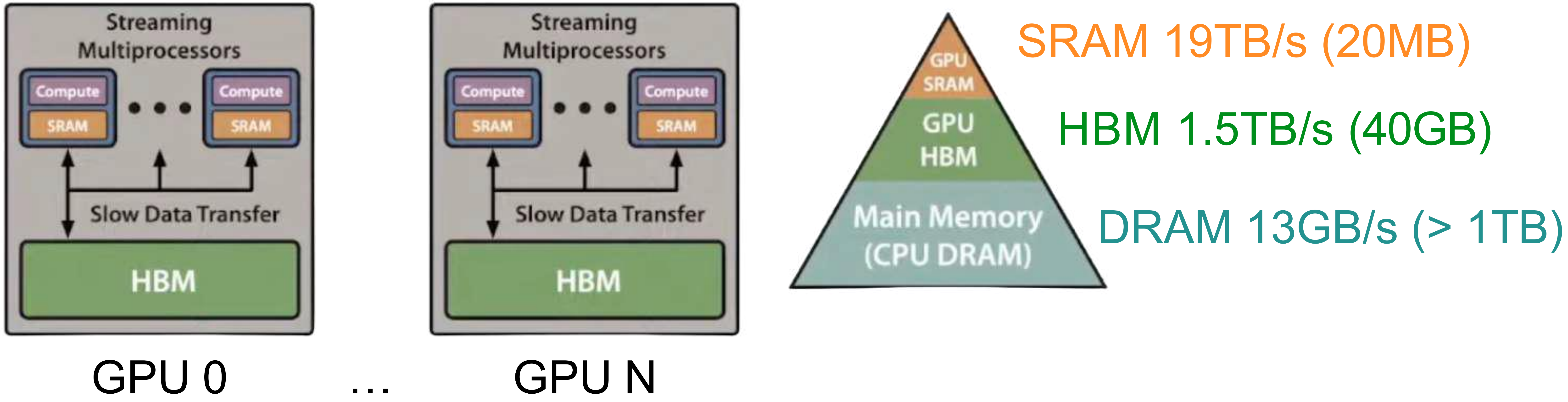
$$\mathbf{O} = \text{Dropout}(\text{Softmax}(\text{Mask}(\mathbf{QK^T})))\mathbf{V}$$

Naive implementation requires repeated R/W from slow GPU HBM. Hard to scale to long sequences



Source: https://arxiv.org/abs/2205.14135

Flash attention



Source: https://arxiv.org/abs/2205.14135

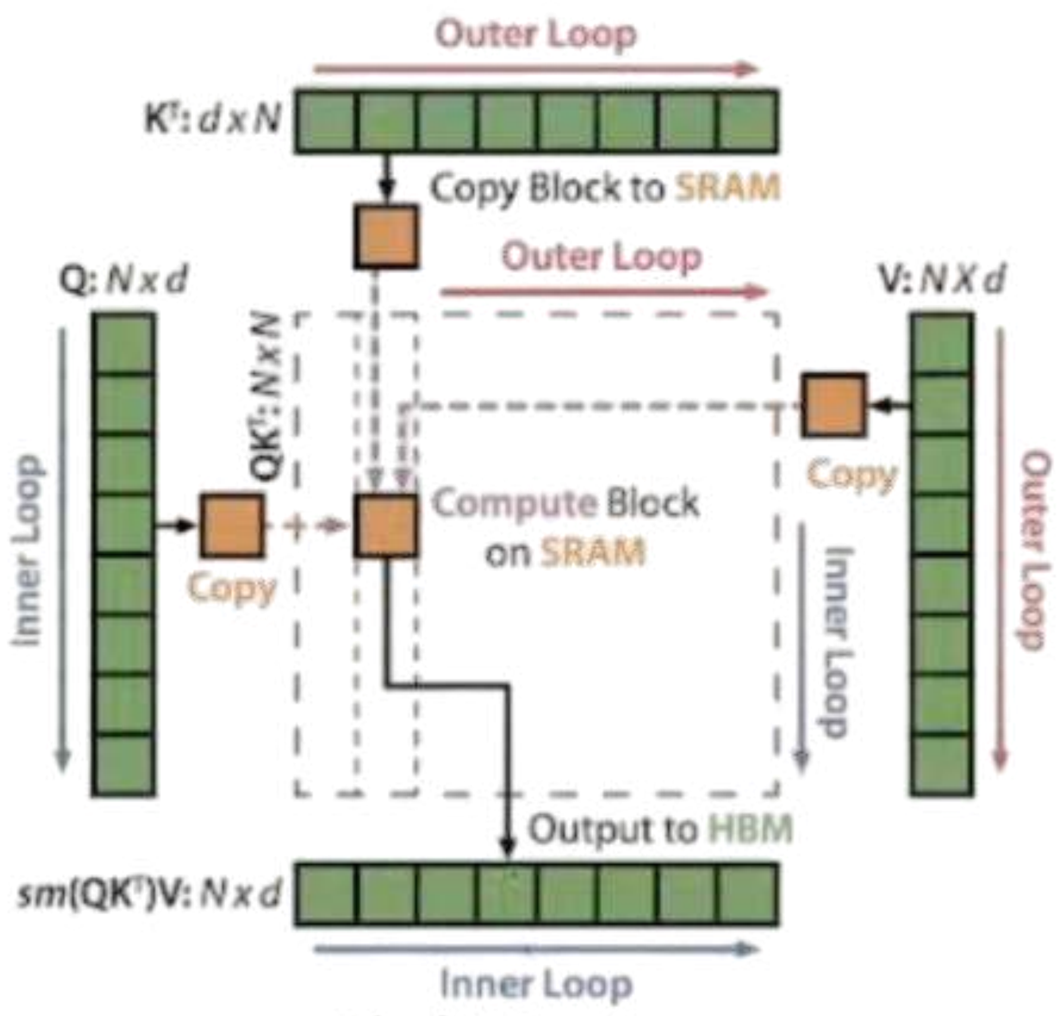
Flash attention

Decomposing large softmax into smaller ones by scaling.

$$\text{softmax}([A_1, A_2]) = [\alpha\ \text{softmax}(A_1), \beta\ \text{softmax}(A_2)]$$

$$\text{softmax}([A_1, A_2]) \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \alpha\ \text{softmax}(A_1)\ V_1 + \ \beta\ \text{softmax}(A_2) V_2$$

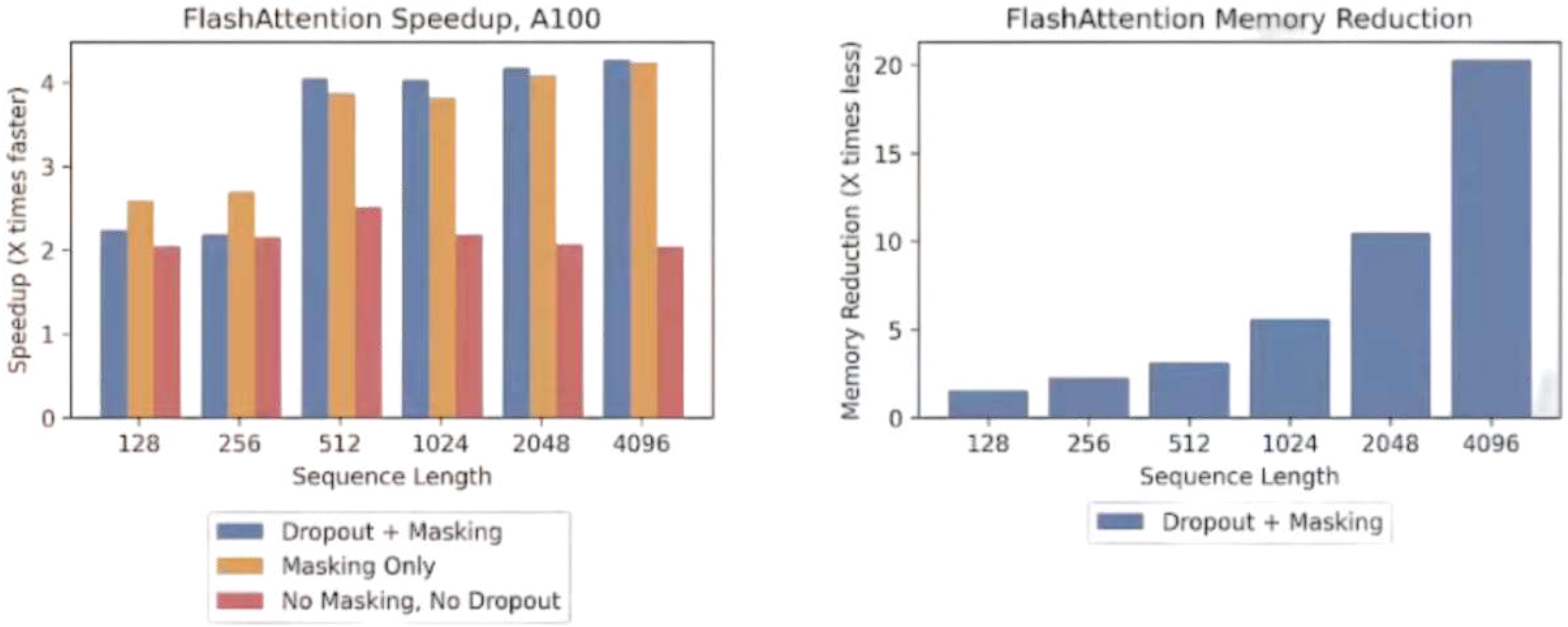
1. Load inputs by blocks from HBM to SRAM.
2. On chip, compute attention output wrt that block.
3. Update output in HBM by scaling.



FlashAttention

Source: https://arxiv.org/abs/2205.14135

Flash attention

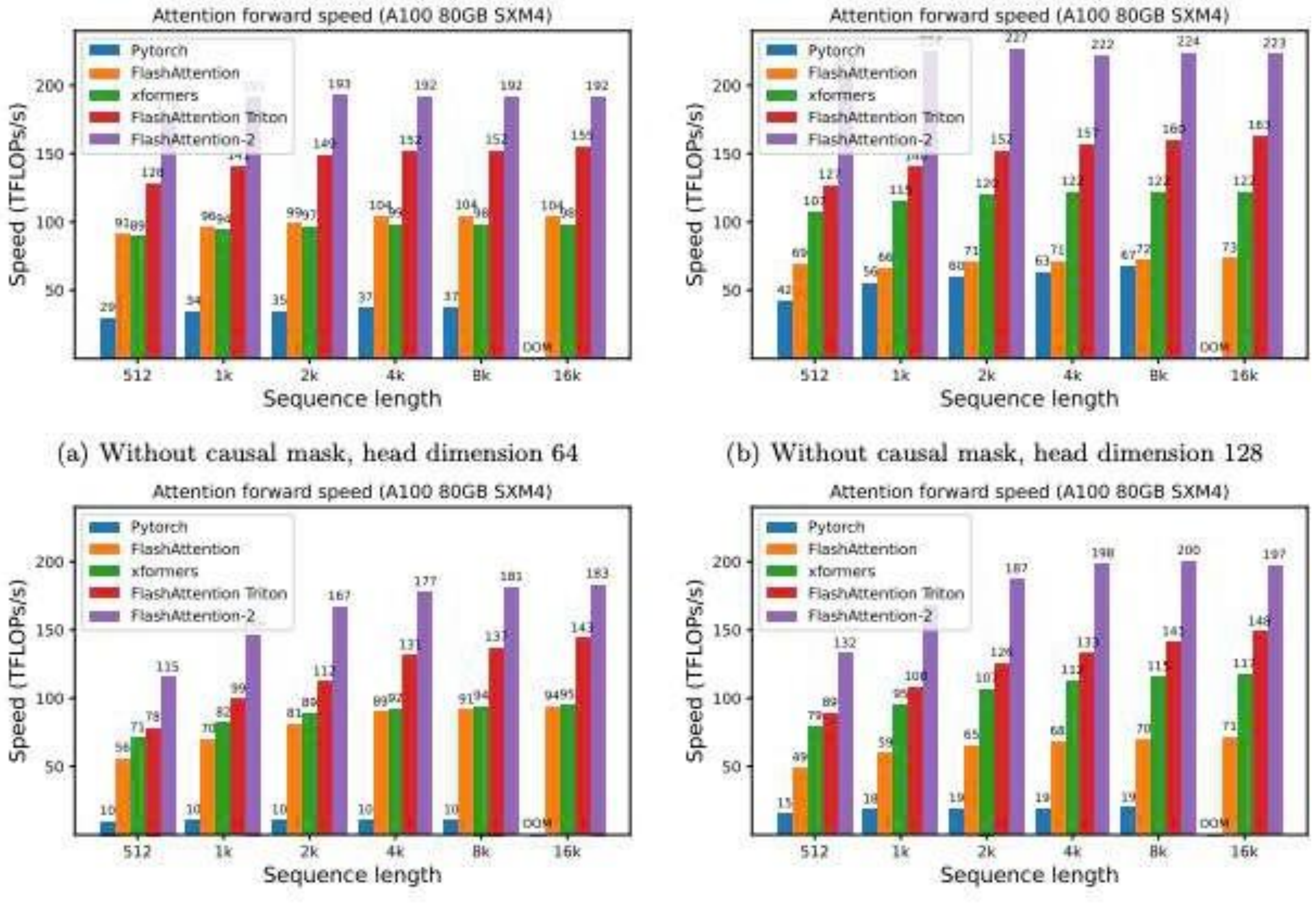


2-4x speedup — with no approximation!

10-20x memory reduction — memory linear in sequence length

Source: https://arxiv.org/abs/2205.14135

Flash attention 2



(a) Without causal mask, head dimension 64

(b) Without causal mask, head dimension 128

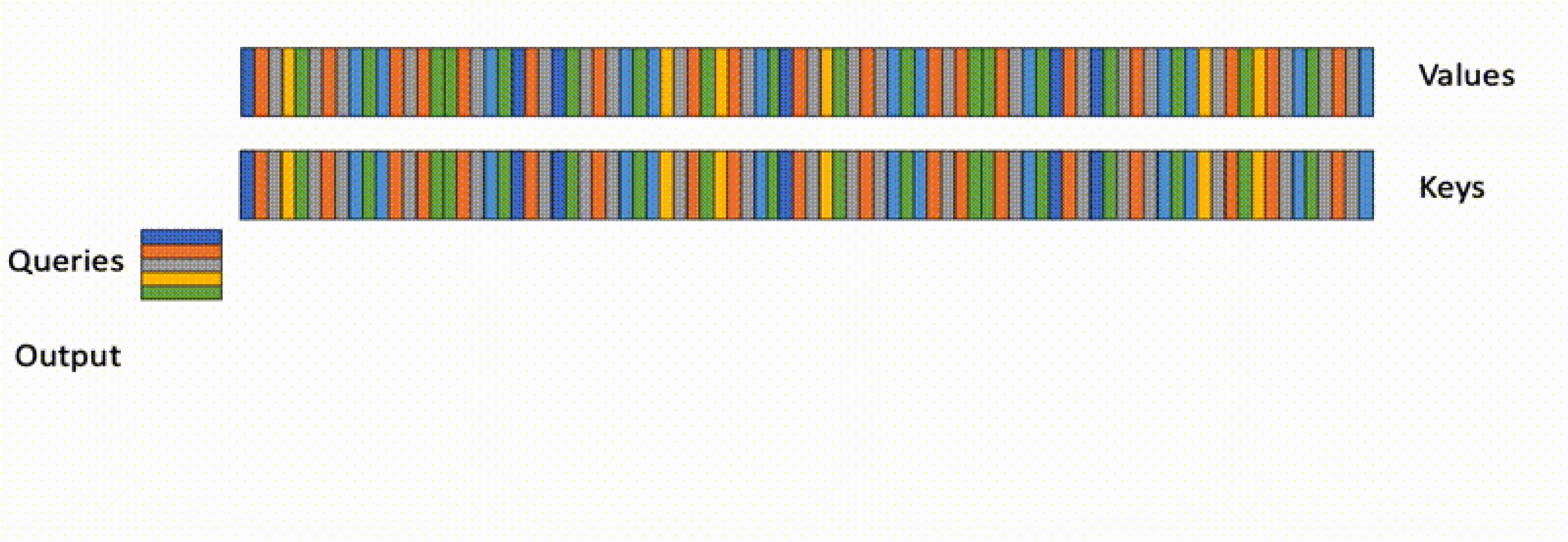
(c) With causal mask, head dimension 64

(d) With causal mask, head dimension 128

Figure 5: Attention forward speed on A100 GPU

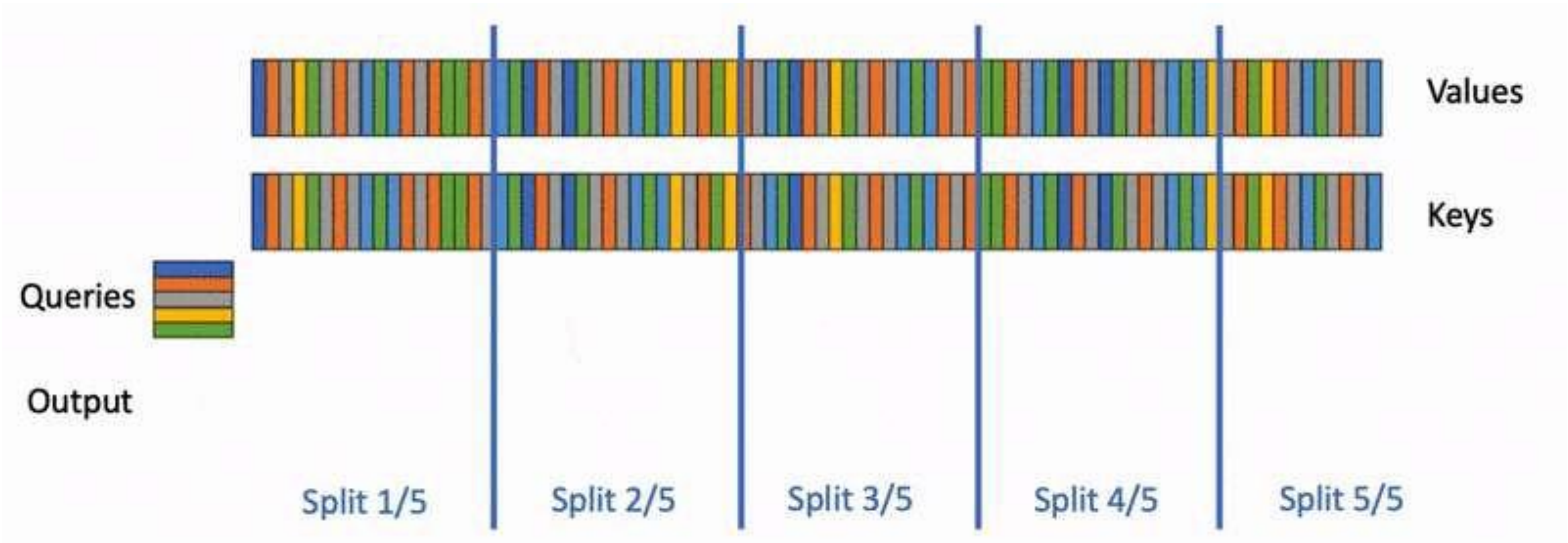
Source: https://arxiv.org/pdf/2307.08691.pdf

Flash decoding



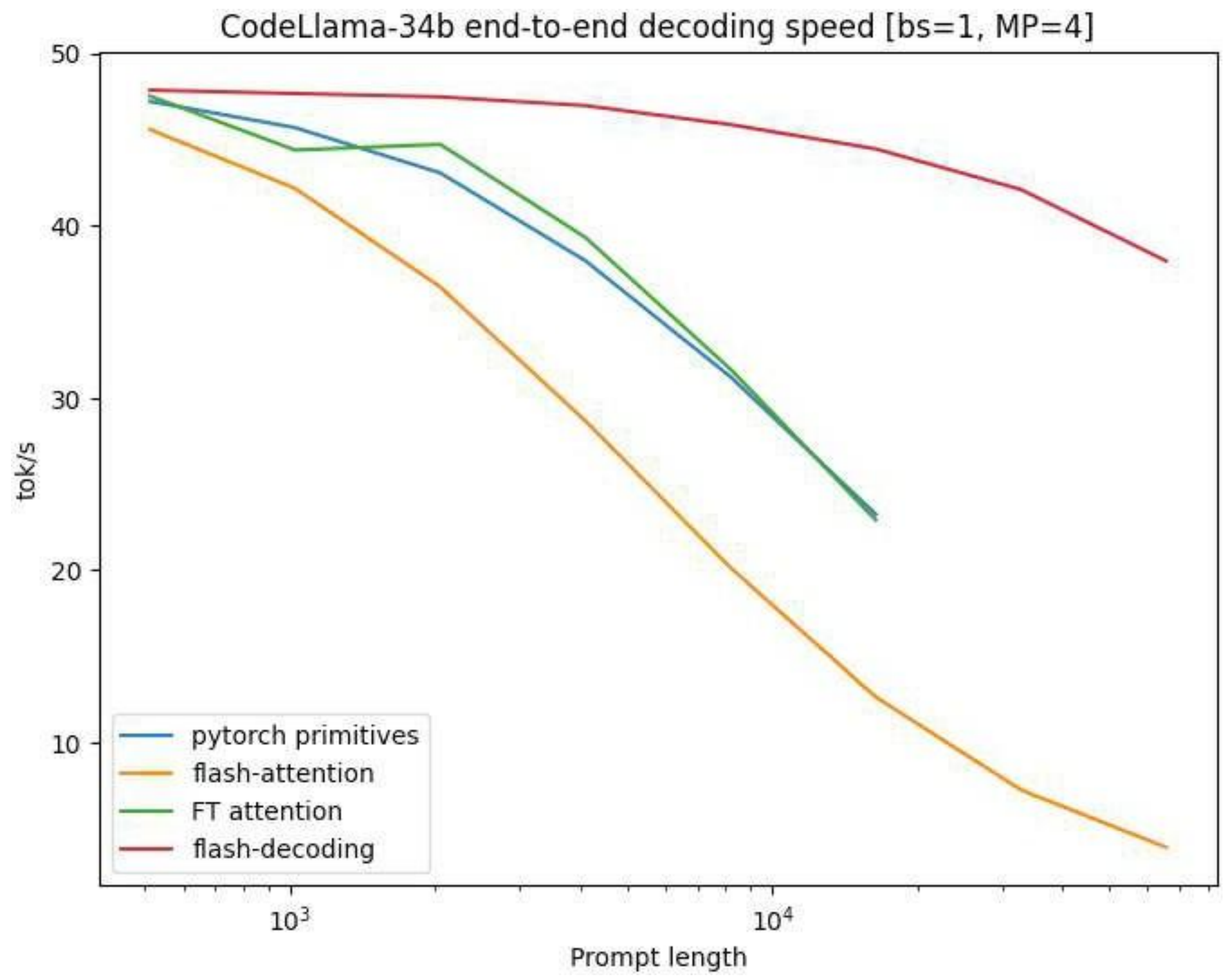
Source: https://pytorch.org/blog/flash-decoding/

Flash decoding



Source: https://pytorch.org/blog/flash-decoding/

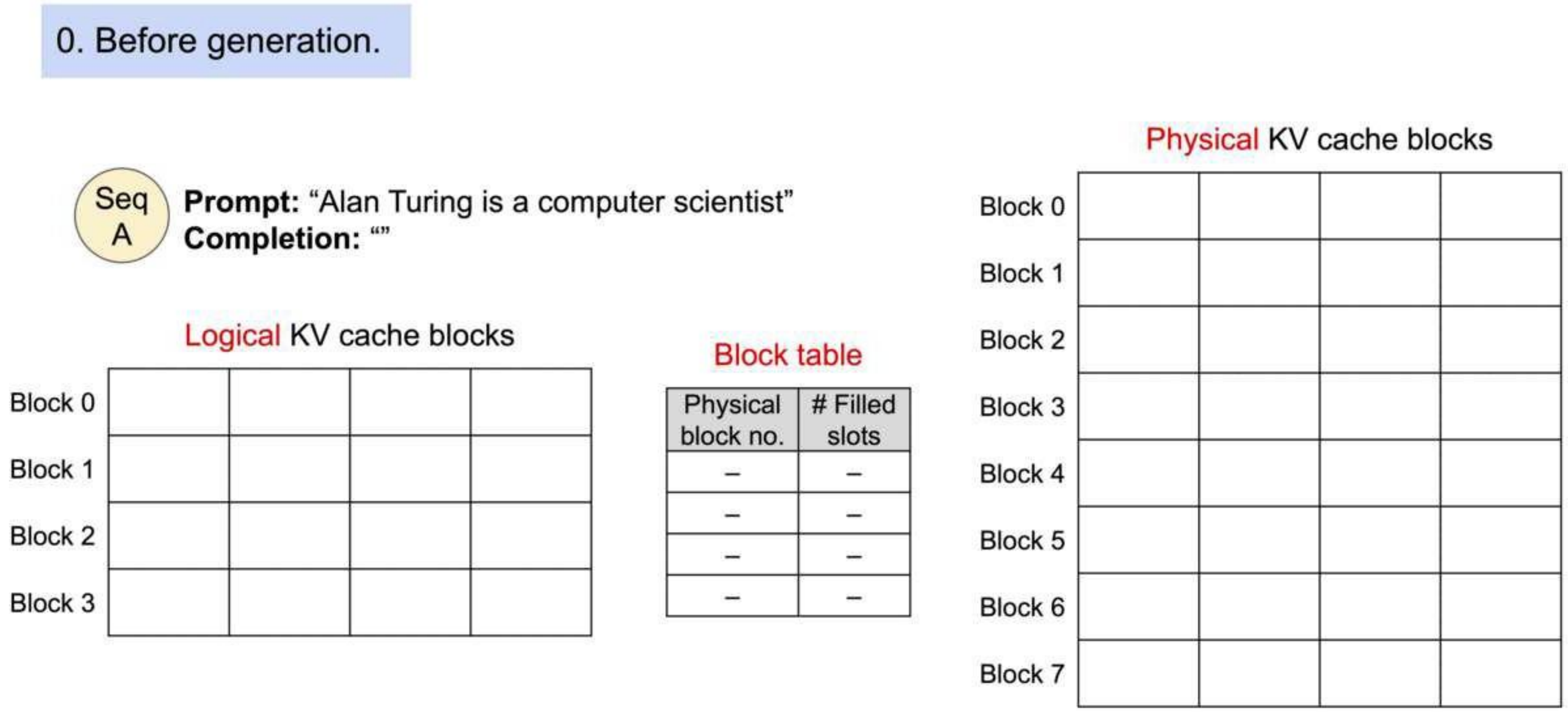
Flash decoding



Source: https://pytorch.org/blog/flash-decoding/

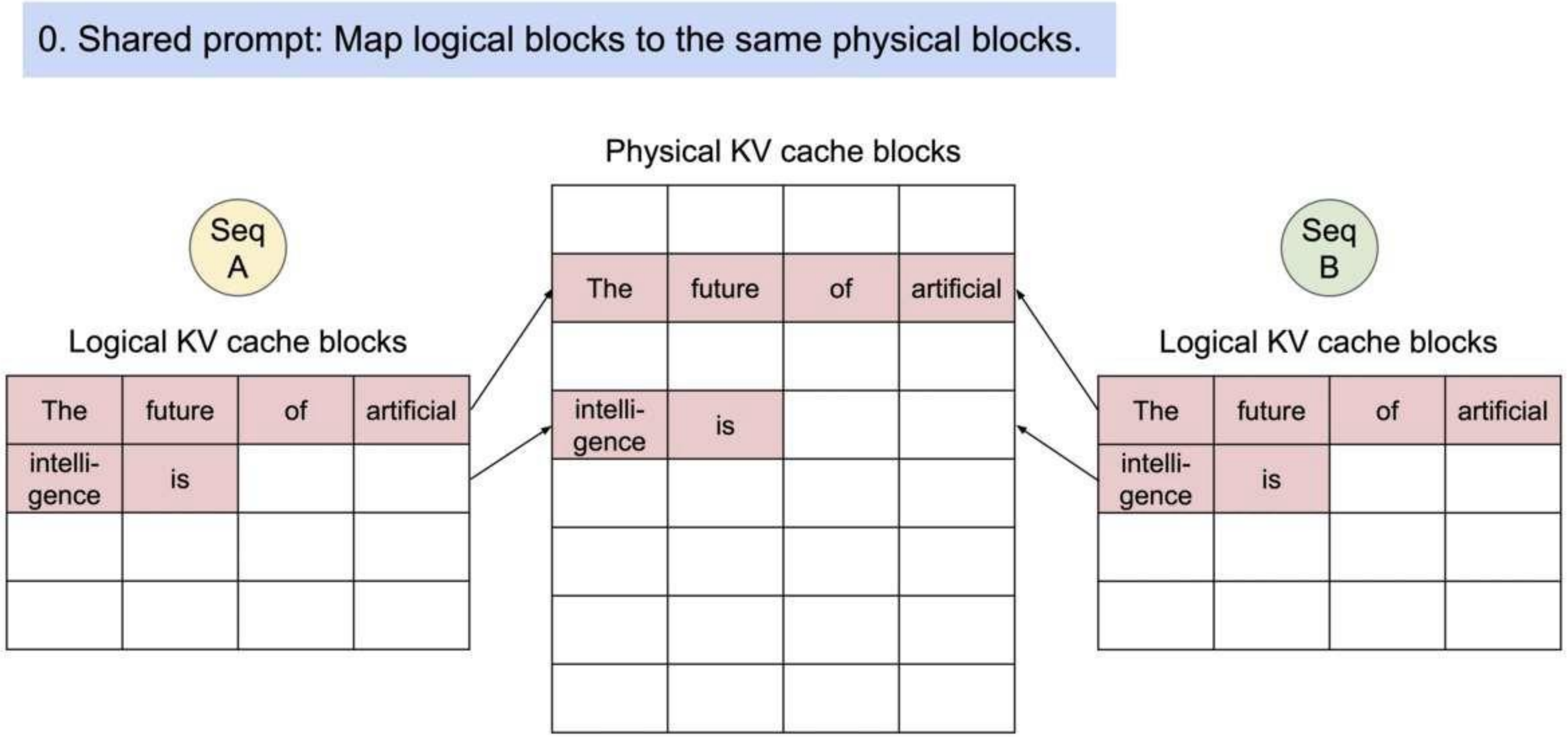
What about long sequence batches?

Paged Attention



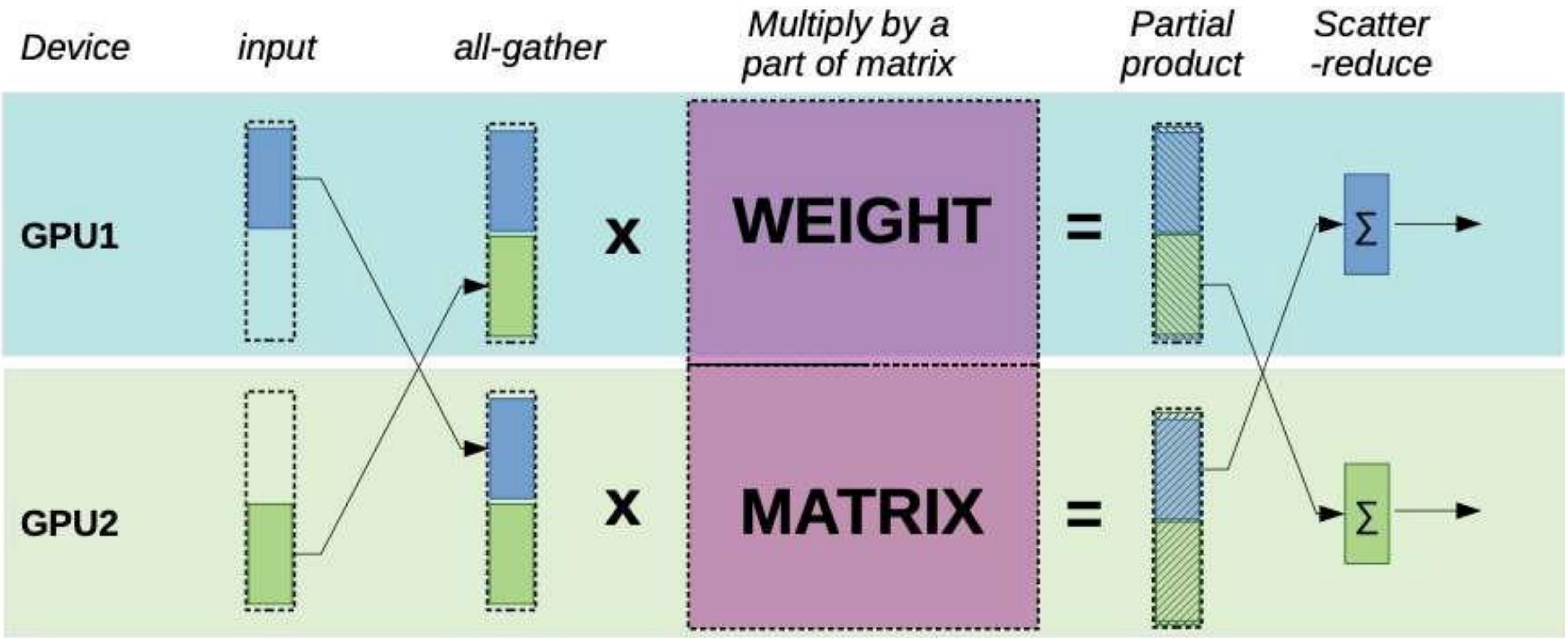
Source: https://blog.vllm.ai/2023/06/20/vllm.html

Paged Attention



Source: https://blog.vllm.ai/2023/06/20/vllm.html

Tensor parallel



Tensor parallel

Performance of GPT-20B

Batch_size	Input Length	Output Length	Latency of single GPU (ms)	Latency of 2-way TP (ms)	Latency of 4-way TP (ms)	Latency of 8-way TP (ms)
1	20	8	225	147	102	89
2	20	8	225	152	108	94
4	20	8	228	158	113	100
8	20	8	239	169	121	107
16	20	8	268	191	133	113
32	20	8	331	230	155	127
64	20	8	452	314	200	169
128	20	8	726	484	318	256
256	20	8	1352	844	533	416
1	60	20	560	358	248	212
2	60	20	562	378	262	222
4	60	20	582	393	274	236
8	60	20	635	429	299	247
16	60	20	748	510	345	272
32	60	20	933	620	418	325
64	60	20	1352	887	574	454
128	60	20	2218	1384	928	699
256	60	20	4141	2424	1574	1152

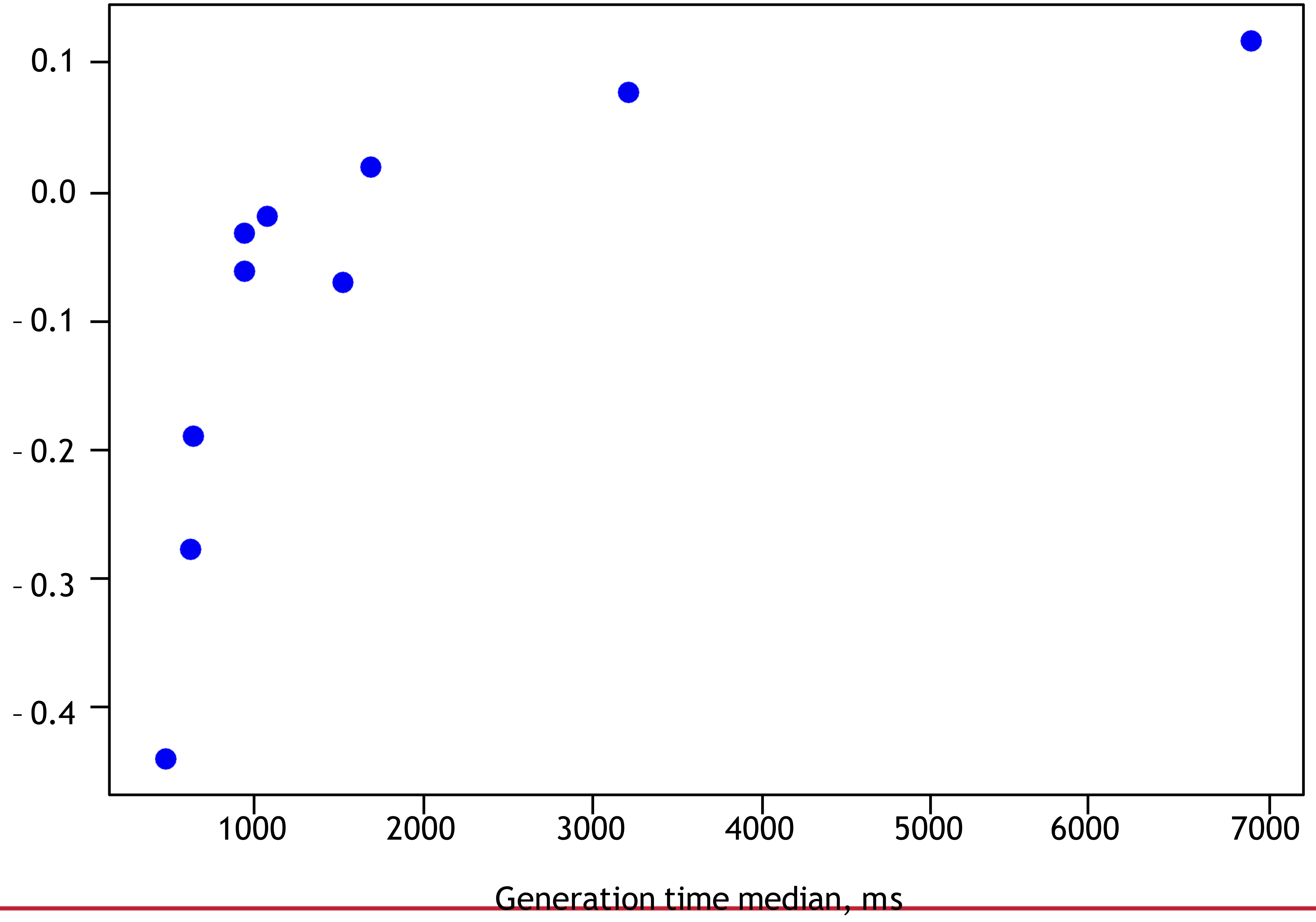
Frameworks

- https://github.com/vllm-project/vllm
https://github.com/InternLM/lmdeploy
https://github.com/microsoft/DeepSpeed-MII
https://github.com/NVIDIA/TensorRT-LLM

https://github.com/ggerganov/llama.cpp

Pareto Curve: Quality vs Compute

Latency/RPS on the X-axis
Quality on the Y-axis



Compute bound vs Memory bound

1024 x 1024 **X** 1024 x 1024

Throughput = Y

1024 x 1024 **X** 1 x 1024

Throughput?

1024*Y?

Compute bound vs Memory bound

1024 x 1024 X 1024 x 1024

FLOP = 10^9

Bandwidth = 2 * 10^6

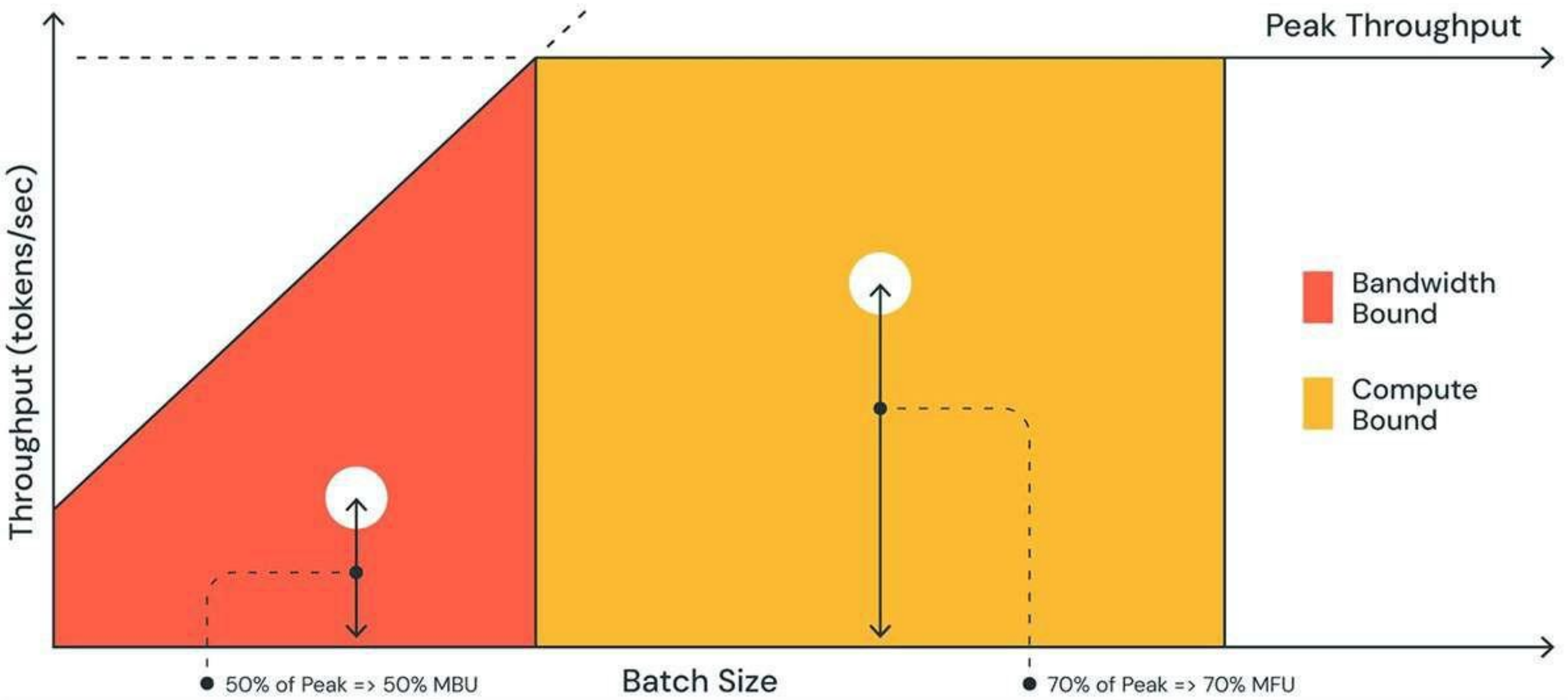
1024 x 1024 X 1 x 1024

FLOP = 10^6

Bandwidth = 10^6

FLOPS vs Bandwidth utilisation

MBU = (achieved memory bandwidth) / (peak memory bandwidth)
achieved memory bandwidth = (total model parameter size + KV cache size) / TPOT



For example:
7B model in float16 has TPOT = 14ms
14 GB parameters in 14ms => 1TB/s bandwidth
peak bandwidth = 2TB / sec
MBU=50%

Knowledge Distillation General framework

- **Teacher p(y|x)**: usually a LLM, e.g. GPT-3 (175B) achieves SOTA quality
- **Student q(y|x)**: small LM, e.g. T5 XL (3B) unable to rich teacher's quality by ordinary training
- **Knowledge Distillation (KD)**: process of teaching the student to imitate teacher's performance

$$L(p(y|x), q_\theta(y|x)) \to \min_\theta$$

Hard-label Distillation

- Sample pairs from teacher
- Finetune model on that pairs

$$\mathop{\mathbb{E}}_{p(y)} \log q_\theta(y) \to \max_\theta$$

$$y^{(1)}, \ldots, y^{(N)} \sim p(y)$$

$$\frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T_n} \log q_\theta(y_t^{(n)} | y_{<t}^{(n)})$$

Soft-label Distillation

- Same as hard-label, but reproduce logits
- Sample from teacher by default
- Hack: use targets from original dataset and compute logits in parallel

$$\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T_n}\sum_{v\in\mathcal{V}} p(y_t^{(n)} = v \mid y_{<t}^{(n)}) \log q_\theta(y_t^{(n)} = v \mid y_{<t}^{(n)})$$

KL Distillation: Idea

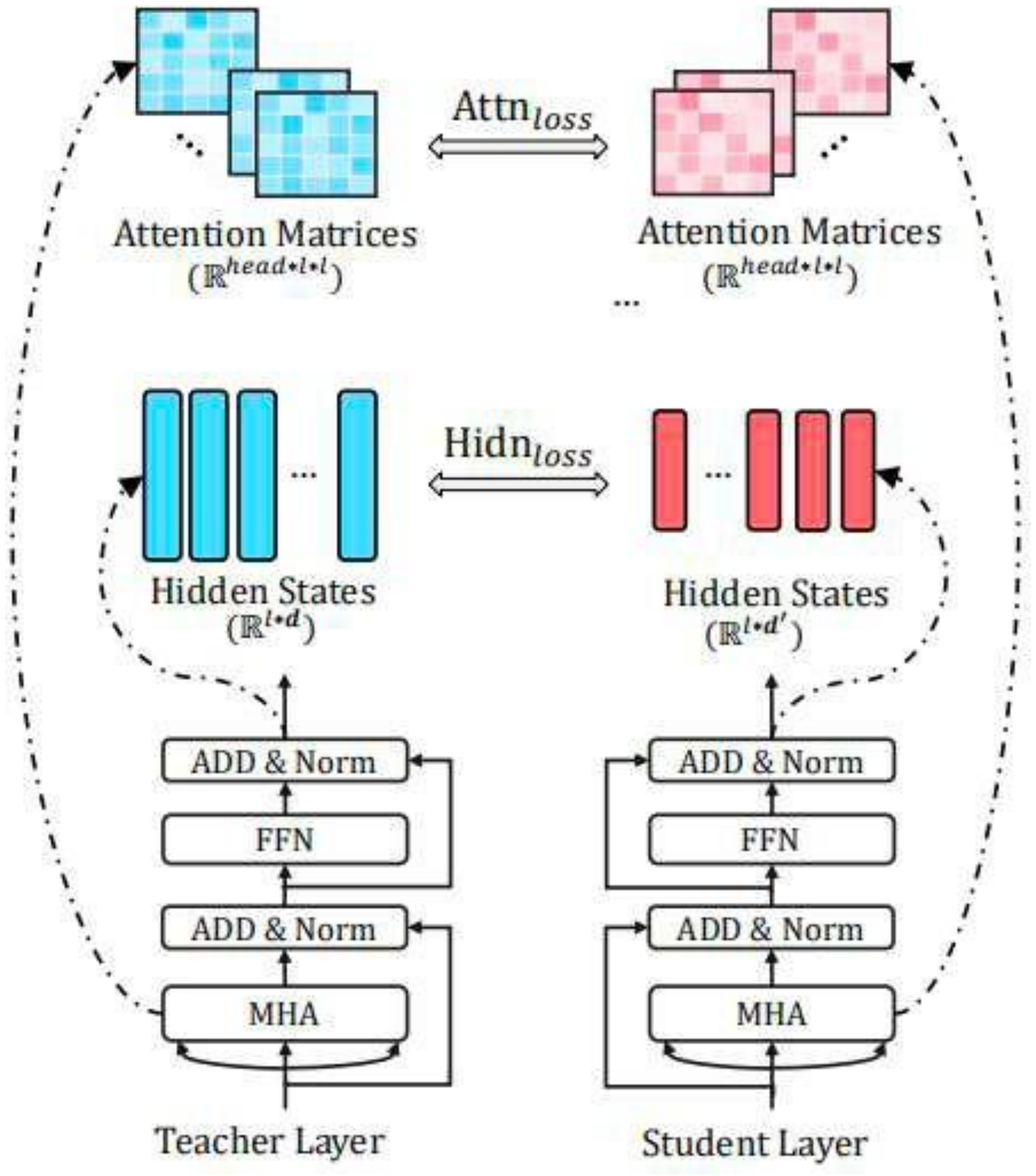
- Distance between distributions
- Monte-Carlo estimation

$$D_{\mathrm{KL}}(p(y) \,||\, q_\theta(y)) \to \min_\theta$$

$$\mathop{\mathbb{E}}_{p(y)} \log \frac{p(y)}{q_\theta(y)} = -\mathop{\mathbb{E}}_{p(y)} \log q_\theta(y) + \mathrm{const} \to \min_\theta$$

Tiny Bert

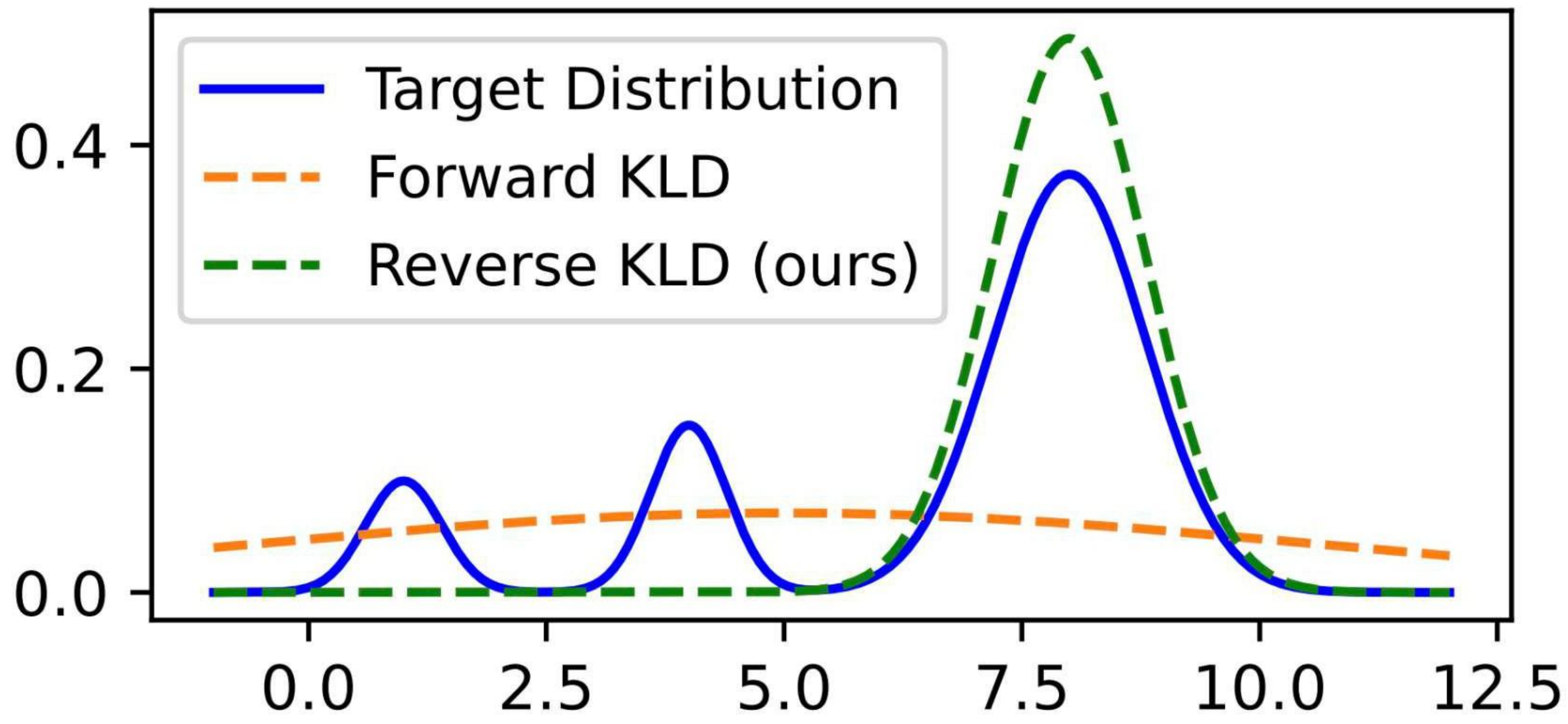
- $L_{attn} = \frac{1}{h} \sum \; MSE(A_i^S, A_i^T)$
- $L_{hidden} = MSE(H^S W_h, H^T)$
- $L_{embed} = MSE(E^S W_e, E^T)$
- $L_{pred} = CE(\frac{z^T}{t}, \frac{z^S}{t})$
- H^S, H^T - хиддены студента и учителя
- A^S, A^T - аттеншн мапы
- z - выходы модели



KL Distillation: LLM issue

- Student's distribution should cover teacher's distribution
- While naturally student is less expressive than teacher

$$D_{\mathrm{KL}}(p(y) \,||\, q_\theta(y)) = \mathop{\mathbb{E}}_{p(y)} \log \frac{p(y)}{q_\theta(y)}$$



Reverse KL

- Swap arguments of KL
- Student approx. only the top probs

$$D_{\mathrm{KL}}(q_\theta(y) \mid\mid p(y)) = \mathop{\mathbb{E}}_{q_\theta(y)} \log \frac{q_\theta(y)}{p(y)}$$

Reverse KL

- Swap arguments of KL
- Student approx. only the top probs
- **Another problem occurs!**
- Unable to differentiate by sampled y

$$D_{\mathrm{KL}}(q_\theta(y) \,||\, p(y)) = \mathop{\mathbb{E}}_{q_\theta(y)} \log \frac{q_\theta(y)}{p(y)}$$

Reverse KL

$$\tilde{q}_\theta(y_t \mid y_{<t}) = \alpha p(y_t \mid y_{<t}) + (1-\alpha) q_\theta(y_t \mid y_{<t})$$

$$R_{t+1}^{\text{Norm}} = \frac{1}{T-t-1} \sum_{k=t}^{T} \log \frac{q_\theta(y_k \mid y_{<k})}{p(y_k \mid y_{<k})}$$

- Importance sampling
- Regularisation term
- Length normalisation

$$\nabla_\theta \mathcal{L} = \mathop{\mathbb{E}}_{\tilde{q}_\theta(y)} \sum_{t=1}^{T} w_t \left[R_{t+1}^{\text{Norm}} \nabla_\theta \log q_\theta(y_t \mid y_{<t}) \right.$$
$$\left. + \nabla_\theta \mathop{\mathbb{E}}_{q_\theta(y_t \mid x, y_{<t})} \log \frac{q_\theta(y_t \mid y_{<t})}{p(y_t \mid y_{<t})} \right]$$

SLIM: Alternative

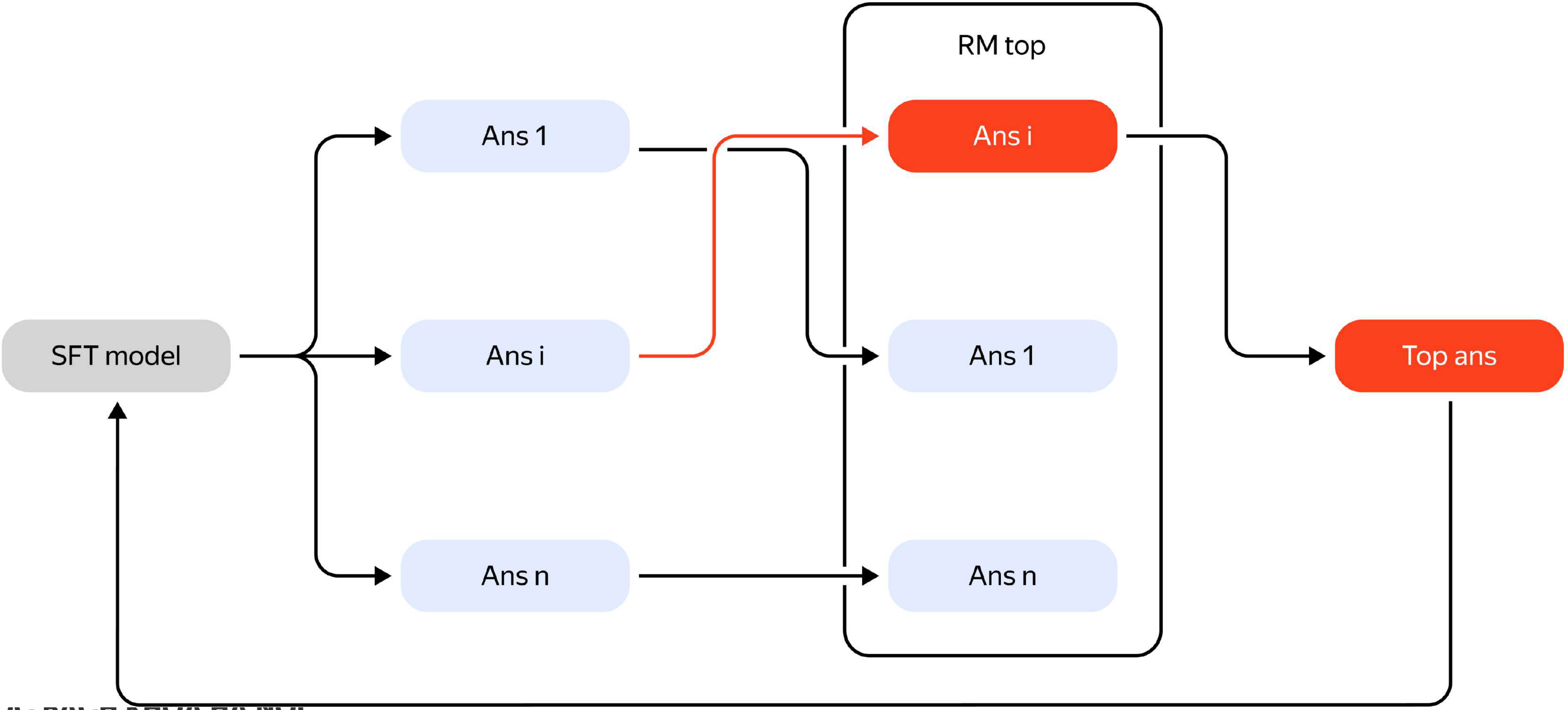
- Take only top %5 logits
- Compatibility with CERL(behaviour cloning)

$$L_{final} = L_{ce} + \alpha(1 - exp(-\frac{s_i}{t_i}))L_{kd}$$

	LLaMA					
	DollyEval		VicunaEval		SelfInst	
	Rouge-L	GPT-4	Rouge-L	GPT-4	Rouge-L	GPT-4
Teacher (LLaMA 13B)	29.7	85.3	19.4	66.3	23.4	76.2
SFT	26.3	79.47	17.5	61.5	20.8	70.6
MiniLLM	28.7	82	19.8	63.8	20.5	72.1
SLIM (Ours)	**29.2**	**82.6**	**20.1**	**64.1**	**23.2**	**73.2**
	LLaMA 2					
	DollyEval		VicunaEval		SelfInst	
	Rouge-L	GPT-4	Rouge-L	GPT-4	Rouge-L	GPT-4
Teacher (LLaMA 2 13B)	30.2	88.9	21.3	69.3	25.1	79.1
SFT	26.5	80.3	18.3	62.8	21.3	73.4
SLIM (Ours)	**29.3**	**84.6**	**19.9**	**67.1**	**23.4**	**75.1**
	MPT					
	DollyEval		VicunaEval		SelfInst	
	Rouge-L	GPT-4	Rouge-L	GPT-4	Rouge-L	GPT-4
Teacher (MPT-30B-instruct)	44.0	94.7	19.3	67.3	23.5	76.1
SFT	28.6	79.5	16.62	60.7	19.9	70.7
SLIM (Ours)	**31.1**	**83.3**	**17.83**	**63.4**	**22.9**	**73.8**

Table 1: We report the Rouge-L and GPT-4 agreement scores on 3 different datasets across 3 different models. We do not have MiniLLM numbers for LLaMA 2 and MPT experiments since the authors did not open-source their models with these backbones.

CE RL



Experience

- Hard label KD is all you need
- RL as distillation very good

Formally

- FP16 -> INT 1/2/4/8
- VRAM and latency/ rps boost
- WxAy
- Symmetric vs Asymmetric
- Granularity
- Grid

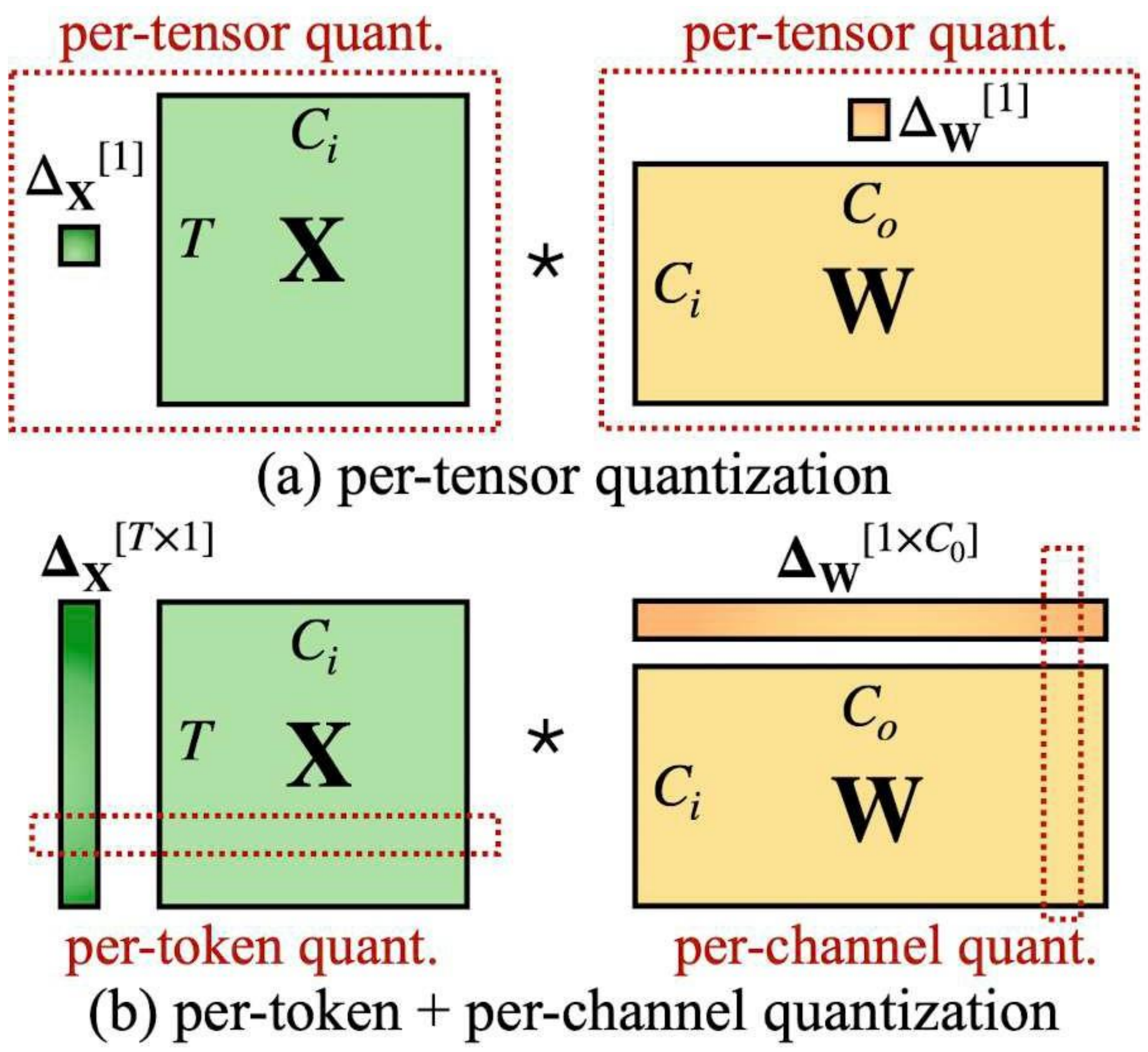
Quantizing a real-valued tensor $\mathbf{x}$ is performed by first mapping it to an unsigned integer grid:

$$\mathbf{x}^{(\mathbb{Z})} = \operatorname{clip}\left(\left\lfloor \frac{\mathbf{x}}{s} \right\rceil + z; 0, 2^b - 1\right), \qquad (1)$$

It is possible to approximately recover the real-valued input $\mathbf{x}$ through an operation that is often referred to as *de-quantization*:

$$\widehat{\mathbf{x}} := q\left(\mathbf{x};\ s, z, b\right) = s\left(\mathbf{x}^{(\mathbb{Z})} - z\right) \approx \mathbf{x}. \qquad (2)$$

In the case of symmetric quantization, we restrict the quantization grid to be symmetric around z.



(a) per-tensor quantization

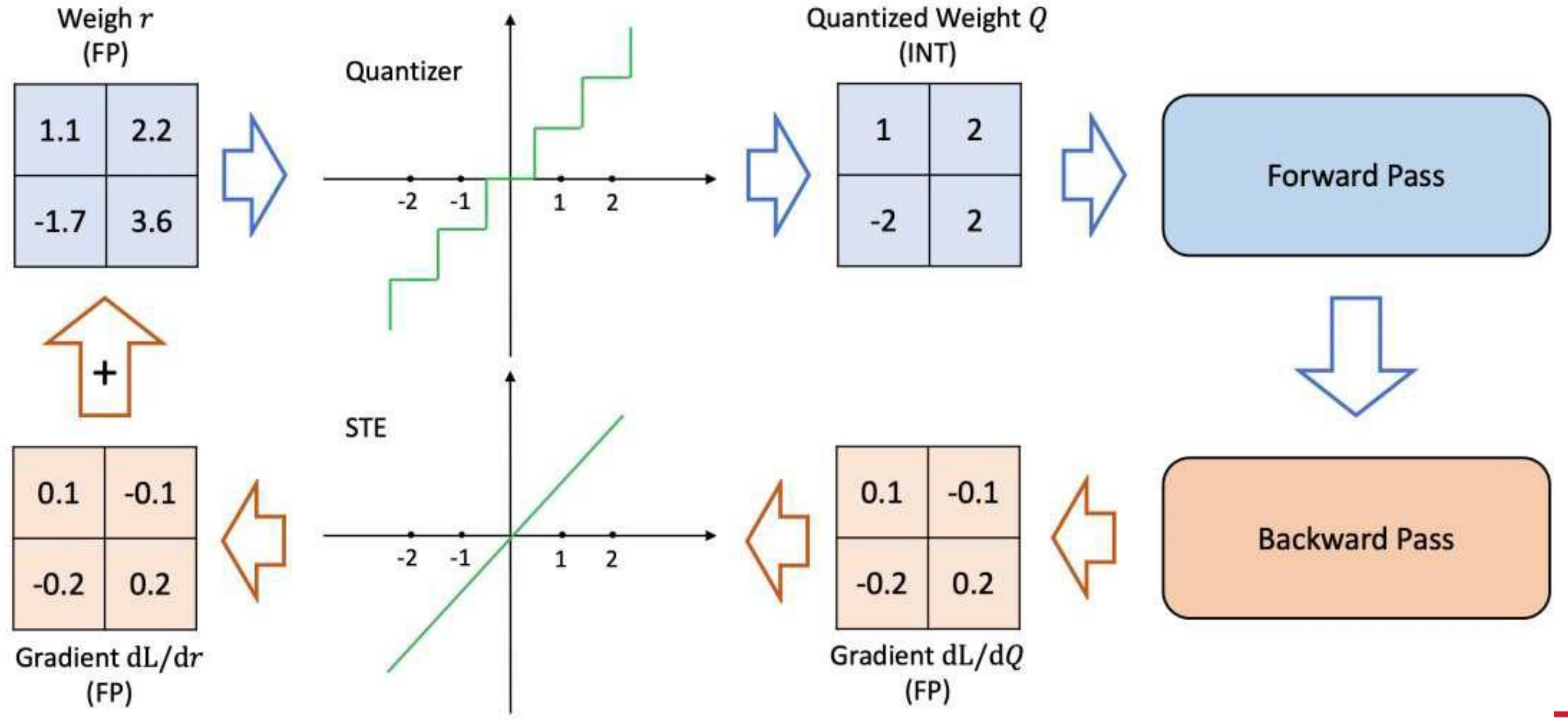
(b) per-token + per-channel quantization

PTQ

- No/~500/1000 calibration samples
- <10 GPU hours
- Calibrating s(scale), z(zero-point)

QAT

- Huge dataset
- Better quaity
- Hard to implement



Quality drop

- Bert example
- Weights - easy
- Activations - hard
- Outliers - problem

Configuration	CoLA	SST-2	MRPC	STS-B	QQP	MNLI	QNLI	RTE	GLUE
FP32	57.27	93.12	88.36	89.09	89.72	84.91	91.58	70.40	**83.06**
W8A8	54.74	92.55	88.53	81.02	83.81	50.31	52.32	64.98	**71.03**
W32A8	56.70	92.43	86.98	82.87	84.70	52.80	52.44	53.07	**70.25**
W8A32	58.63	92.55	88.74	89.05	89.72	84.58	91.43	71.12	**83.23**

Table 1: Post-training quantization results on development sets of the GLUE benchmark (except WNLI). The metrics for these tasks can be found in the GLUE paper (Wang et al., 2018a); in all cases, higher is better. FP32 baseline is trained by the authors from the pre-trained checkpoint, see Appendix B.1 for details. We report a median over 5 runs with different random seeds.

Quantized activations	STS-B	MNLI	QNLI	RTE
none (FP32 model)	89.09	84.91	91.58	70.40
all	62.64	42.67	50.74	48.74
all, except softmax input	70.92	42.54	51.84	48.74
all, except sum of embeddings	67.57	46.82	51.22	51.26
all, except self-attention output	70.47	46.57	50.98	50.90
all, except softmax output	72.83	50.35	50.23	49.46
all, except residual connections after FFN	**81.57**	**82.56**	**89.73**	**67.15**
same as above, but for layers 10, 11 only	**79.40**	**81.24**	**88.03**	**63.90**

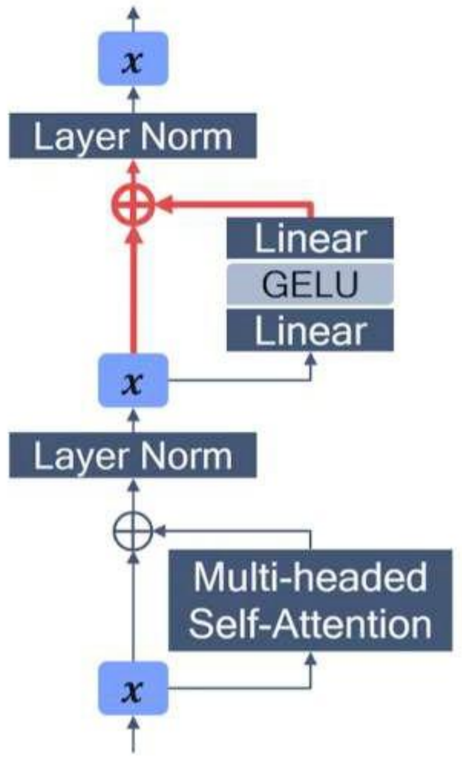
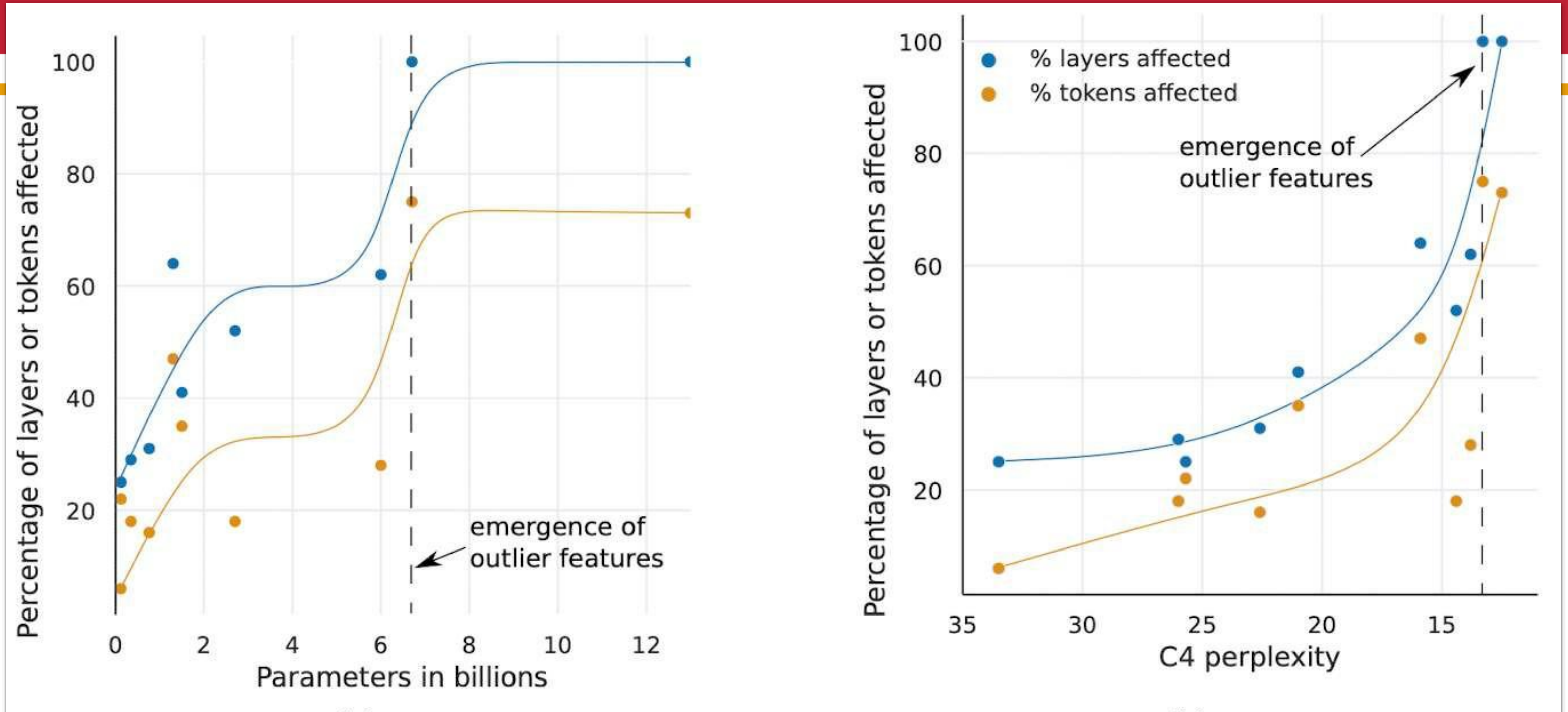


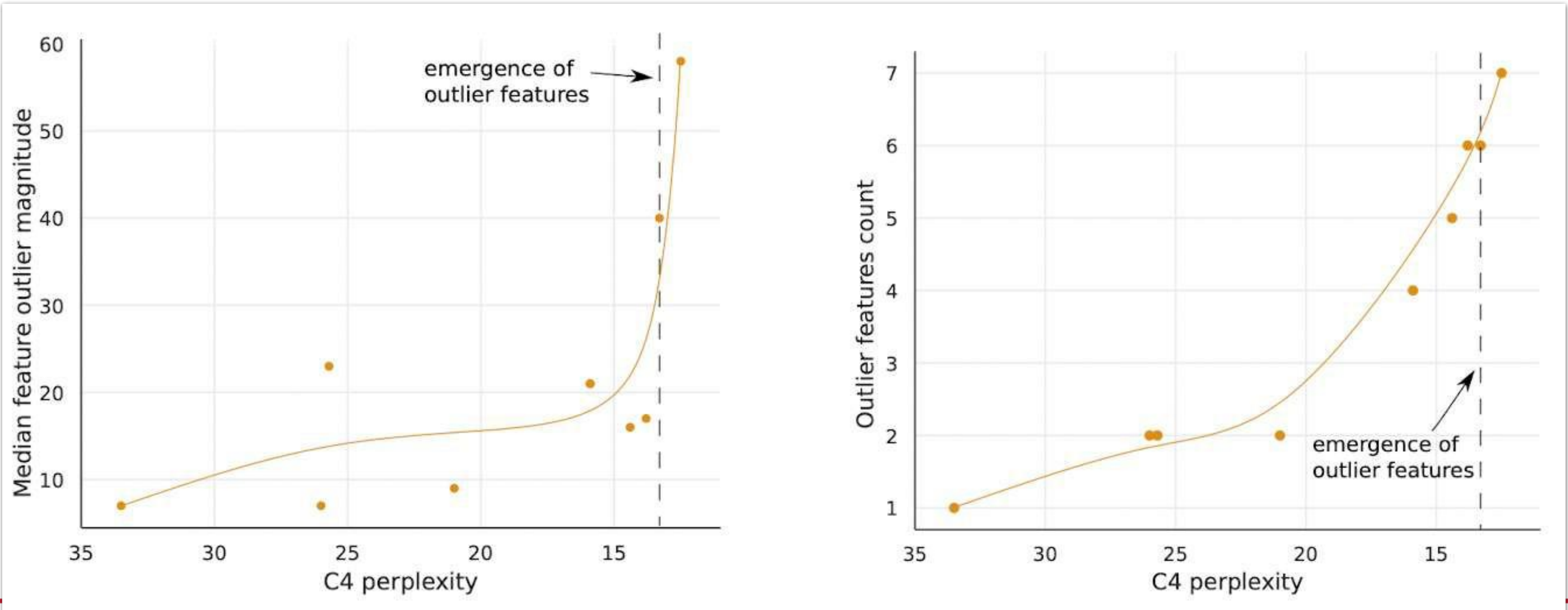
Table 2 & Figure 1: *Left*: Leave-one-out analysis for activation quantizers on problematic GLUE tasks. We set all weights to FP32 and use current min-max (with a batch size of 1) range estimator for activations. We report median score over 5 runs with different random seeds. *Right*: A schematic illustration of the attention layer in BERT. Hidden activation tensor is denoted by **x**. ⊕ is an element-wise addition. A problematic residual connection sum after feed-forward network is highlighted in red.

Outliers is problem

- Outliers occurs in QKVO projections
- Outliers have huge impact on quality



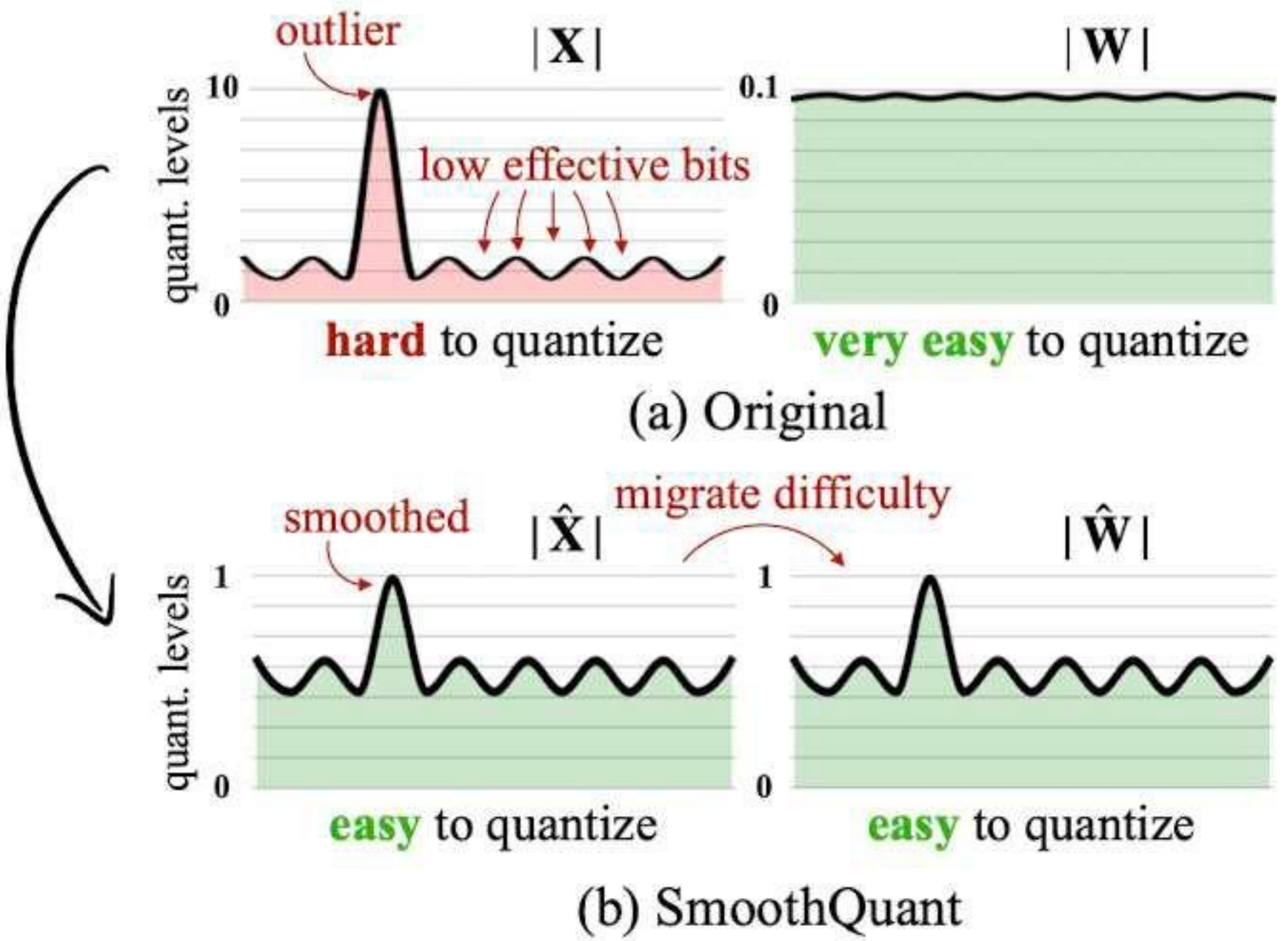
(a) (b)



(a) (b)

Intuition

- Migrate some difficulty to weights

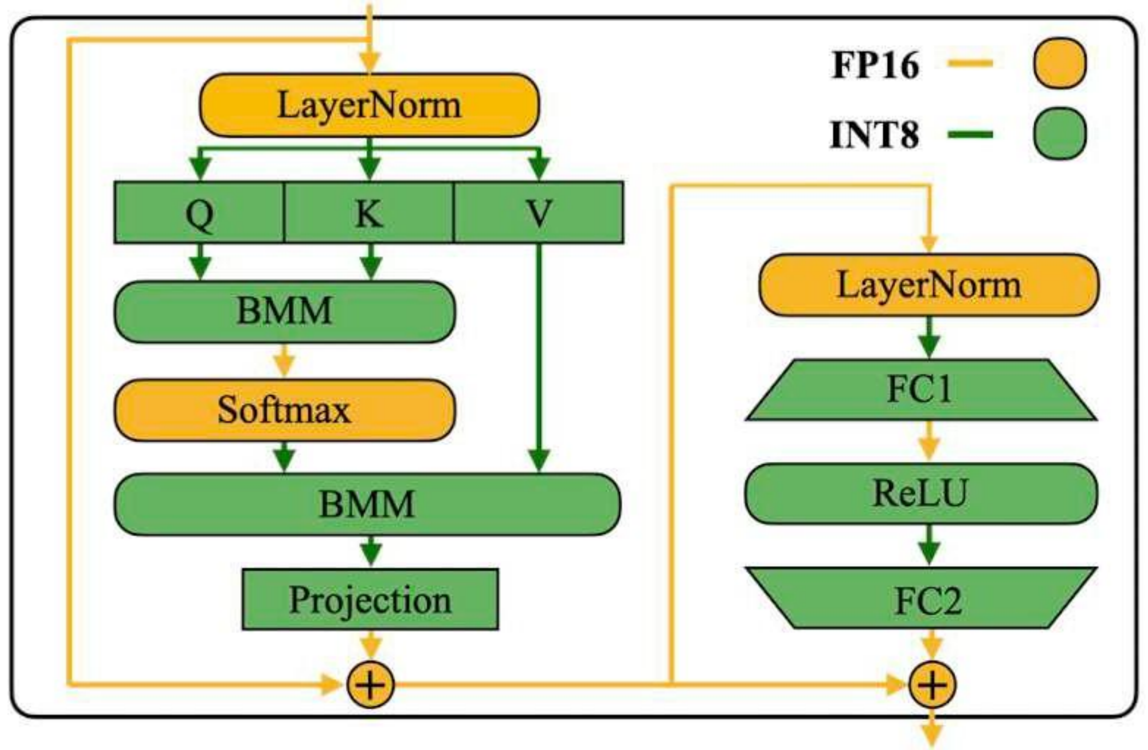


Method

- W8A8 with same quality
- Apply to all BMM
- Fuse scaling with previous operations

Table 2: Among different activation quantization schemes, only per-channel quantization (Bondarenko et al., 2021) preserves the accuracy, but it is *not* compatible (marked in gray) with INT8 `GEMM` kernels. We report the average accuracy on WinoGrande, HellaSwag, PIQA, and LAMBADA.

Model size (OPT-)	6.7B	13B	30B	66B	175B
FP16	64.9%	65.6%	67.9%	69.5%	71.6%
INT8 per-tensor	39.9%	33.0%	32.8%	33.1%	32.3%
INT8 per-token	42.5%	33.0%	33.1%	32.9%	31.7%
INT8 per-channel	64.8%	65.6%	68.0%	69.4%	71.4%

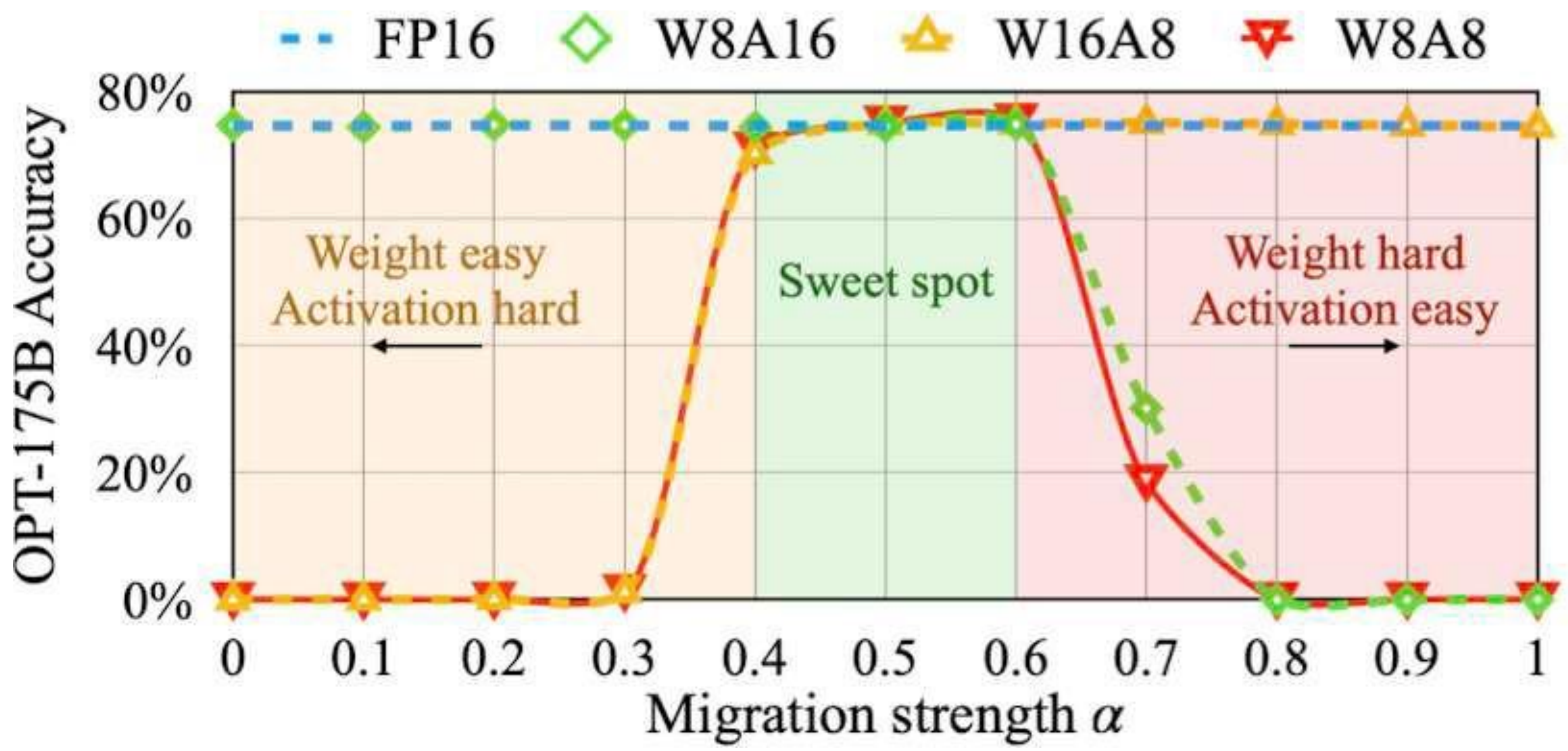


Method

$$X^{INT8} = [\frac{X^{FP16}}{s}], s = \frac{max(|X|)}{127}$$

$$Y = (Xdiag(s)^{-1}) \cdot (diag(s)W)$$

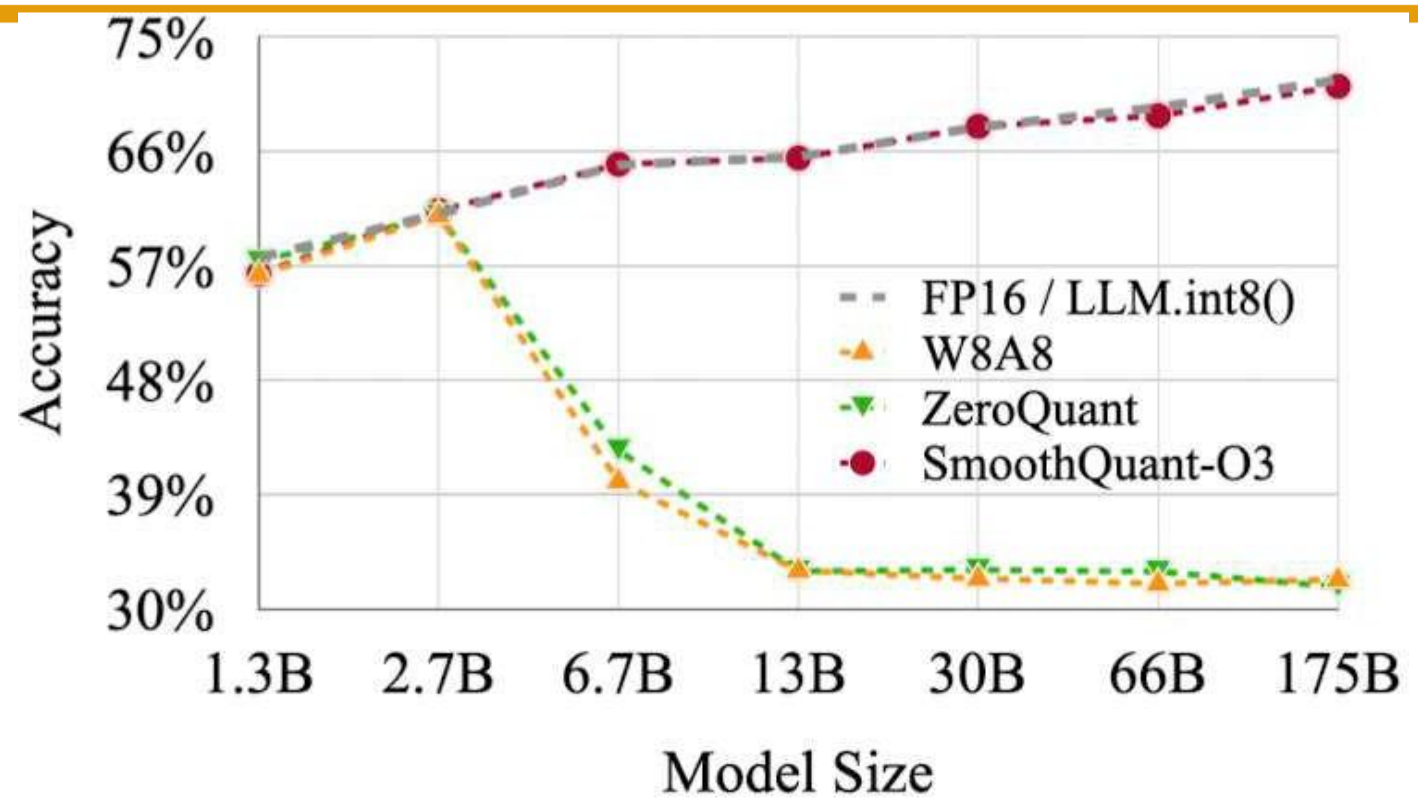
$$s_j = \frac{max(|X_J|)^a}{max(|W_j|)^{1-a}}$$



Speed

Table 8: GPU Latency (ms) of different quantization schemes. The coarser the quantization scheme (from per-token to per-tensor, dynamic to static, O1 to O3, defined in Table 3), the lower the latency. SmoothQuant achieves lower latency compared to FP16 under all settings, while `LLM.int8()` is mostly slower. The batch size is 4.

Model	OPT-13B		OPT-30B	
Sequence Length	256	512	256	512
FP16	152.6	296.3	343.0	659.9
`LLM.int8()`	237.1	371.5	387.9	654.9
SmoothQuant-O1	124.5	243.3	246.7	490.7
SmoothQuant-O2	120.5	235.1	240.2	478.3
SmoothQuant-O3	112.1	223.1	227.6	458.4



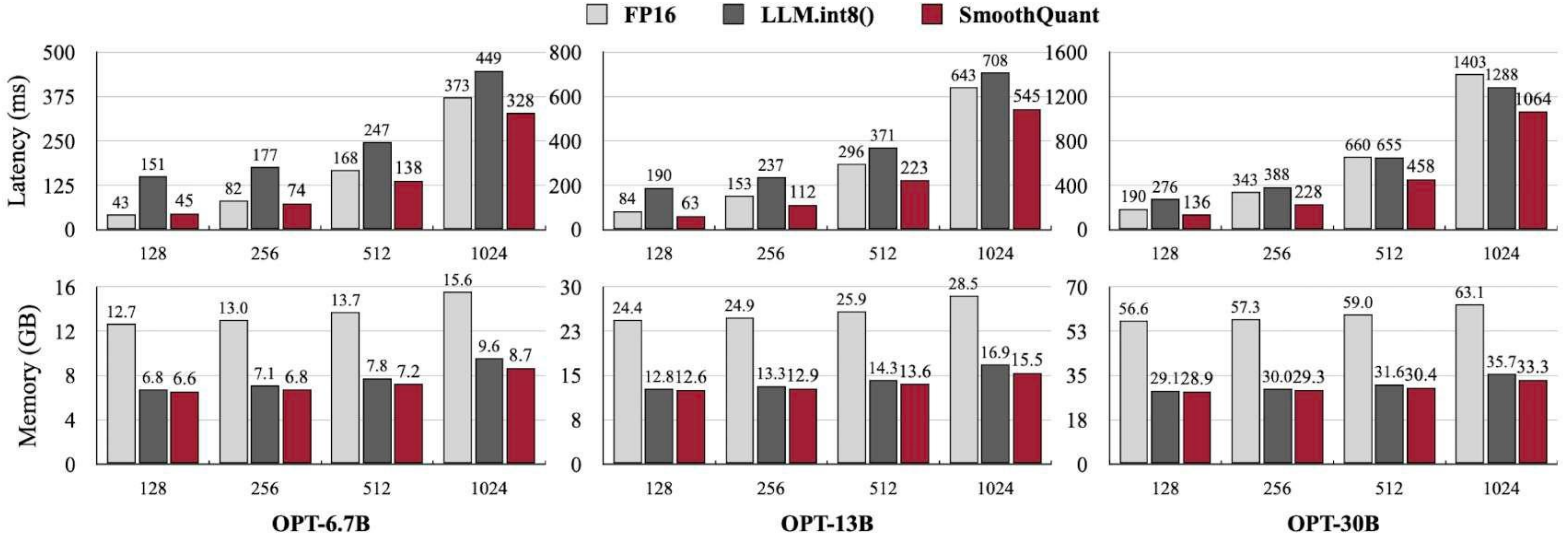


Figure 7: The PyTorch implementation of SmoothQuant-O3 achieves up to **1.51**× speedup and **1.96**× memory saving for OPT models on a single NVIDIA A100-80GB GPU, while `LLM.int8()` slows down the inference in most cases.

W4A16: GPTQ

GPU	FP16	3bit	Speedup	GPU reduction
A6000 – 48GB	589ms	130ms	4.53×	$8 \rightarrow 2$
A100 – 80GB	230ms	71ms	3.24×	$5 \rightarrow 1$

Table 6: Average per-token latency (batch size 1) when generating sequences of length 128.

- Decoding only
- Published code and CUDA kernels
- Solving minimization task layer per layer

$$argmin_{\hat{W}} ||WX - \hat{W}X||_2^2$$

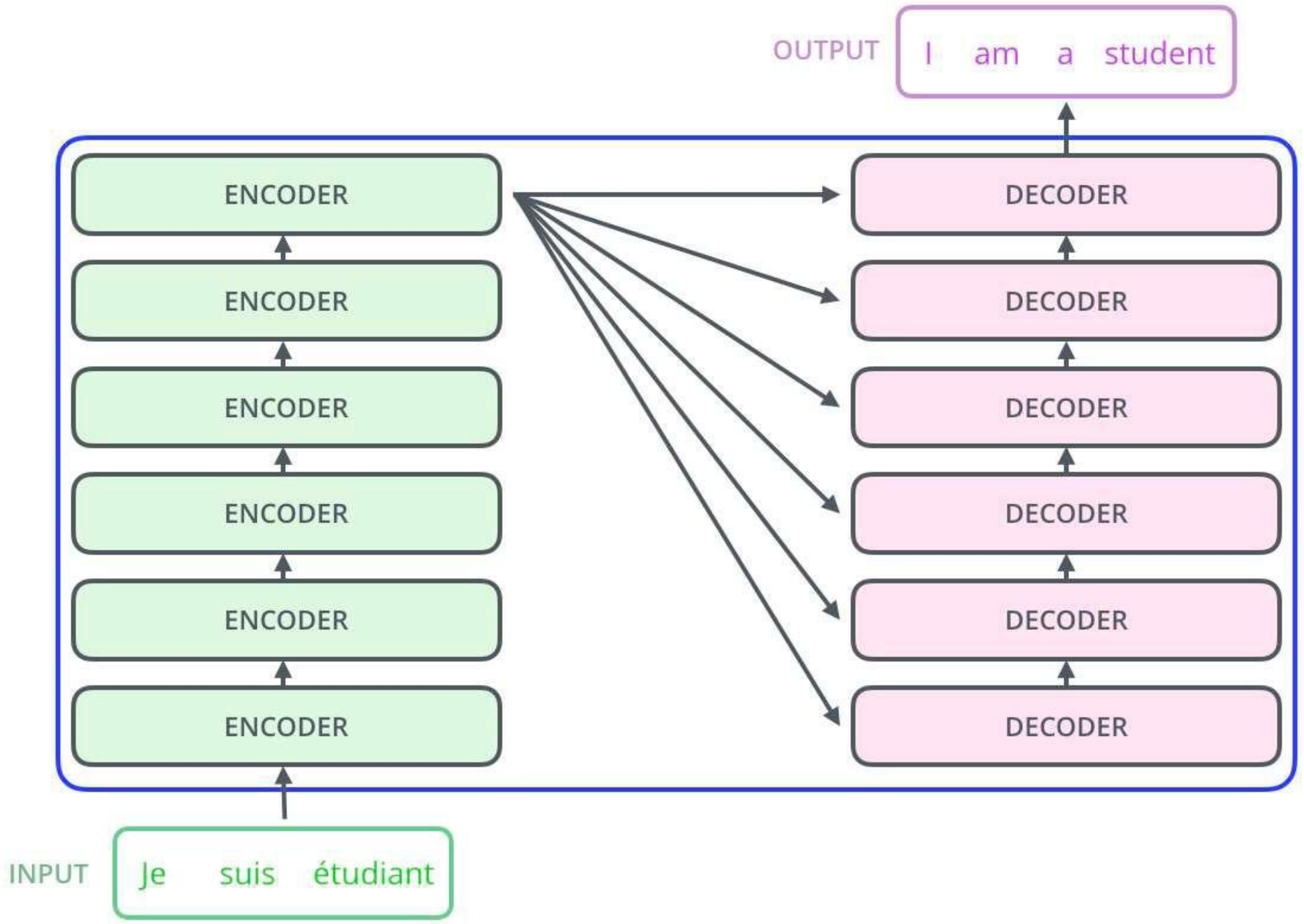
Method	Bits	OPT-175B Wiki2	PTB	C4	LAMB. ↑	BLOOM-176B Wiki2	PTB	C4	LAMB. ↑
Baseline	16	8.34	12.01	10.13	75.59	8.11	14.59	11.71	67.40
RTN	4	10.54	14.22	11.61	71.34	8.37	15.00	12.04	66.70
GPTQ	4	**8.37**	**12.26**	**10.28**	**76.80**	**8.21**	**14.75**	**11.81**	**67.71**
RTN	3	7.3e3	8.0e3	4.6e3	0	571.	107.	598.	0.17
GPTQ	3	**8.68**	**12.68**	**10.67**	**76.19**	**8.64**	**15.57**	**12.27**	**65.10**
GPTQ	3/g1024	8.45	12.48	10.47	77.39	8.35	15.01	11.98	67.47
GPTQ	3/g128	8.45	12.37	10.36	76.42	8.26	14.89	11.85	67.86

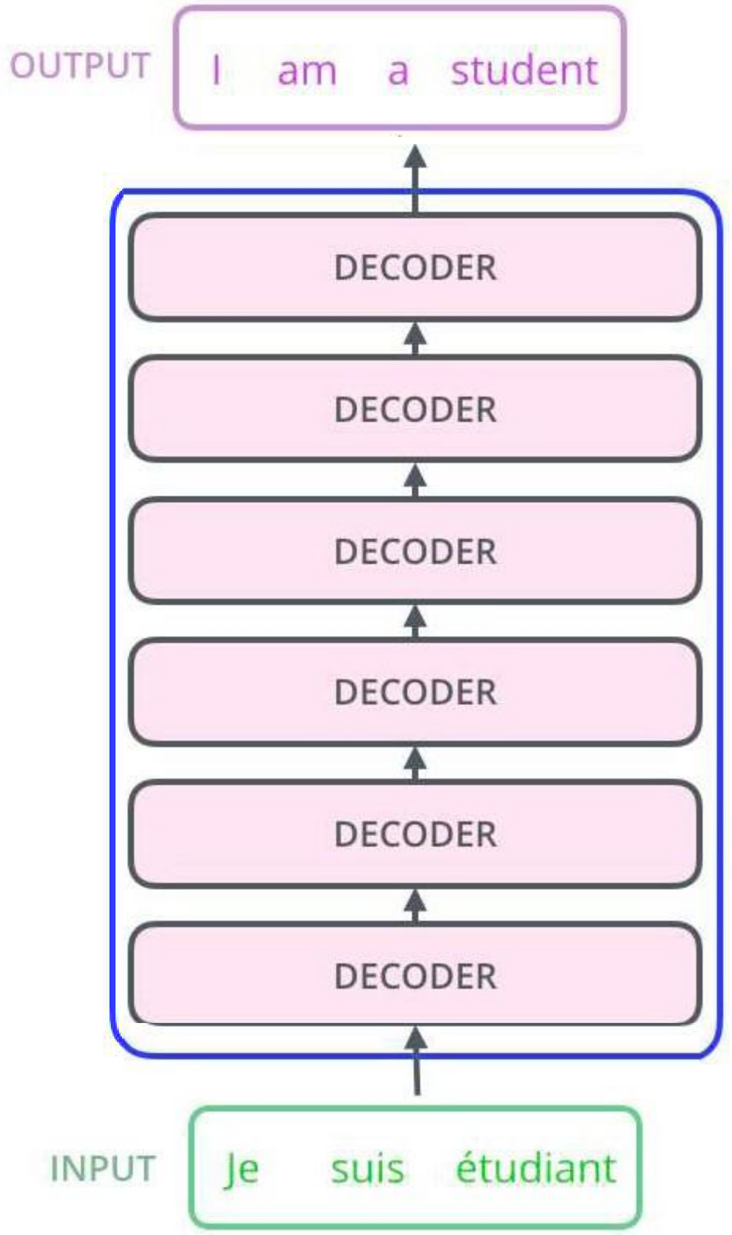
Table 5: Results summary for OPT-175B and BLOOM-176B. “g1024” and “g128” denote results with groupings of size 1024 and 128, respectively.

Architecture

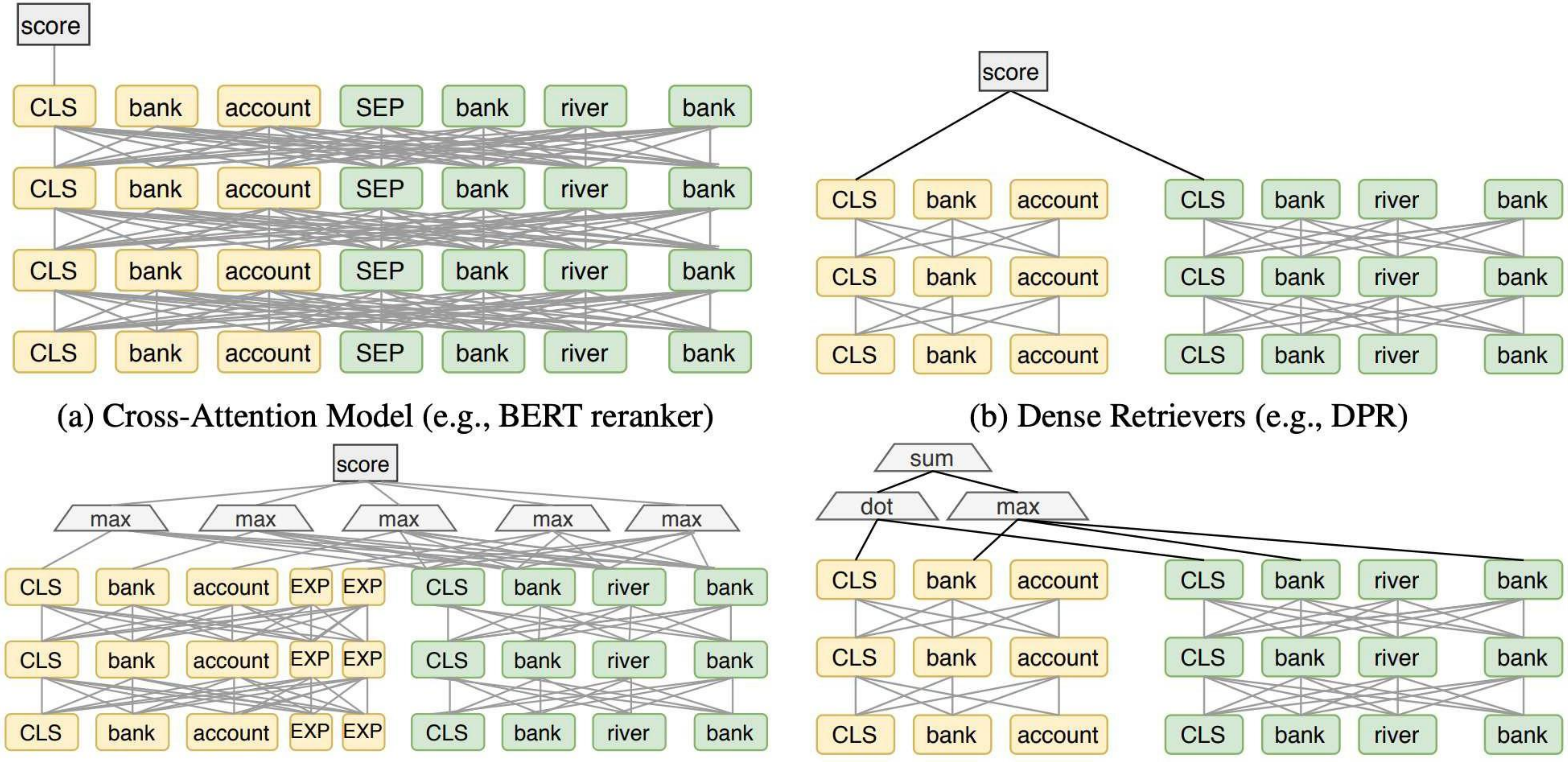
Encoder-decoder vs decoder

Who faster with same amount of parameters ?





Bert tips



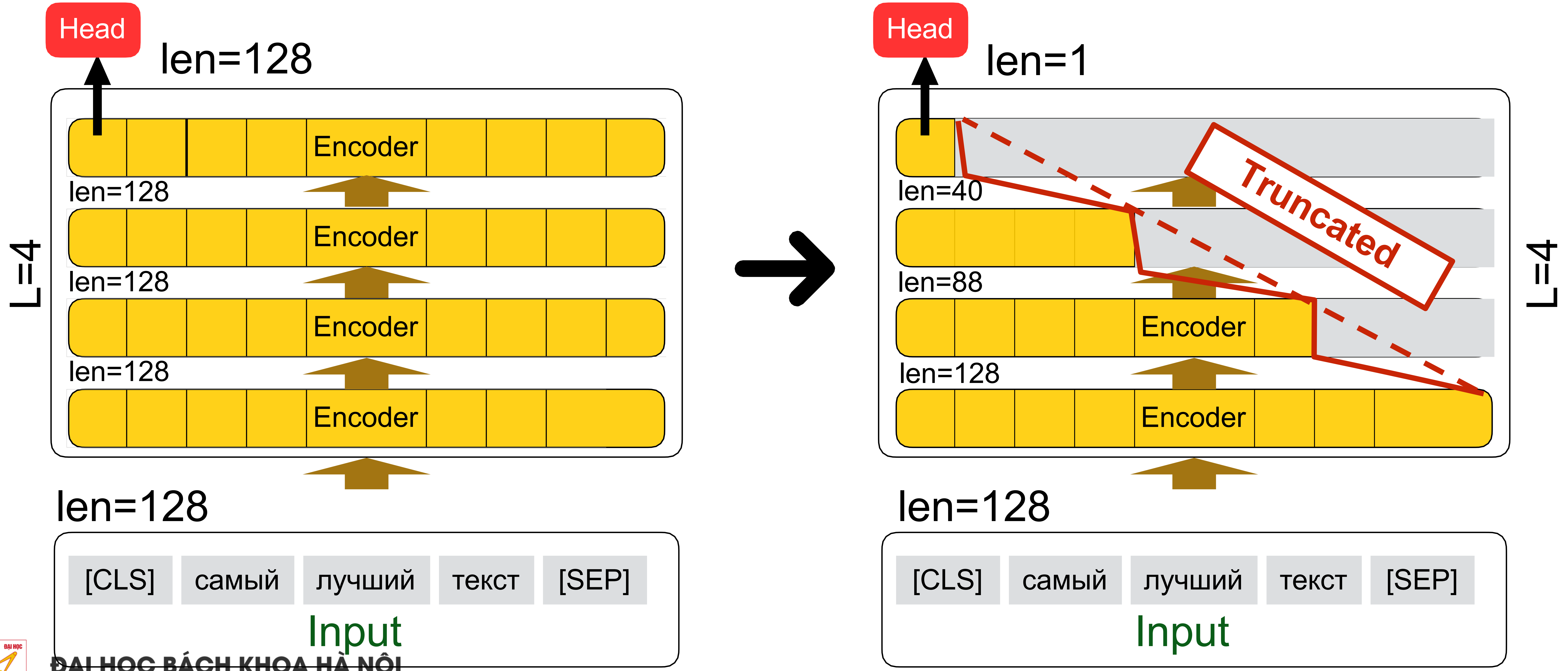
(a) Cross-Attention Model (e.g., BERT reranker)

(b) Dense Retrievers (e.g., DPR)

(c) ColBERT: All-to-All Match

(d) COIL: Contextualized Exact Match

Bert tips


Head
len=128
Encoder
len=128
Encoder
len=128
Encoder
len=128
Encoder
L=4
len=128
[CLS] самый лучший текст [SEP]
Input
Head
len=1
len=40
len=88
Encoder
len=128
Encoder
Truncated
L=4
len=128
[CLS] самый лучший текст [SEP]
Input

Speculative decoding: Vanilla idea

- Small draft model, e.g 7B
- Large verification model, e.g. 70B
- Decode 1 to K+1 tokens per iteration
- Verify by large model

Table 1 | **Chinchilla performance and speed on XSum and HumanEval with naive and speculative sampling at batch size 1 and $K = 4$.** XSum was executed with nucleus parameter $p = 0.8$, and HumanEval with $p = 0.95$ and temperature 0.8.

Sampling Method	Benchmark	Result	Mean Token Time	Speed Up
ArS (Nucleus)	XSum (ROUGE-2)	0.112	14.1ms/Token	1×
SpS (Nucleus)	XSum (ROUGE-2)	0.114	7.52ms/Token	1.92×
ArS (Greedy)	XSum (ROUGE-2)	0.157	14.1ms/Token	1×
SpS (Greedy)	XSum (ROUGE-2)	0.156	7.00ms/Token	2.01×
ArS (Nucleus)	HumanEval (100 Shot)	45.1%	14.1ms/Token	1×
SpS (Nucleus)	HumanEval (100 Shot)	47.0%	5.73ms/Token	2.46×

Algorithm 2 Speculative Sampling (SpS) with Auto-Regressive Target and Draft Models

Given lookahead K and minimum target sequence length T.
Given auto-regressive target model $q(.|.)$, and auto-regressive draft model $p(.|.)$, initial prompt sequence $x_0, \ldots, x_t$.
Initialise $n \leftarrow t$.
while $n < T$ **do**
  for $t = 1 : K$ **do**
    Sample draft auto-regressively $\tilde{x}_t \sim p(x|, x_1, \ldots, x_n, \tilde{x}_1, \ldots, \tilde{x}_{t-1})$
  end for
  In parallel, compute $K + 1$ sets of logits from drafts $\tilde{x}_1, \ldots, \tilde{x}_K$:

$$q(x|, x_1, \ldots, x_n),\ q(x|, x_1, \ldots, x_n, \tilde{x}_1),\ \ldots,\ q(x|, x_1, \ldots, x_n, \tilde{x}_1, \ldots, \tilde{x}_K)$$

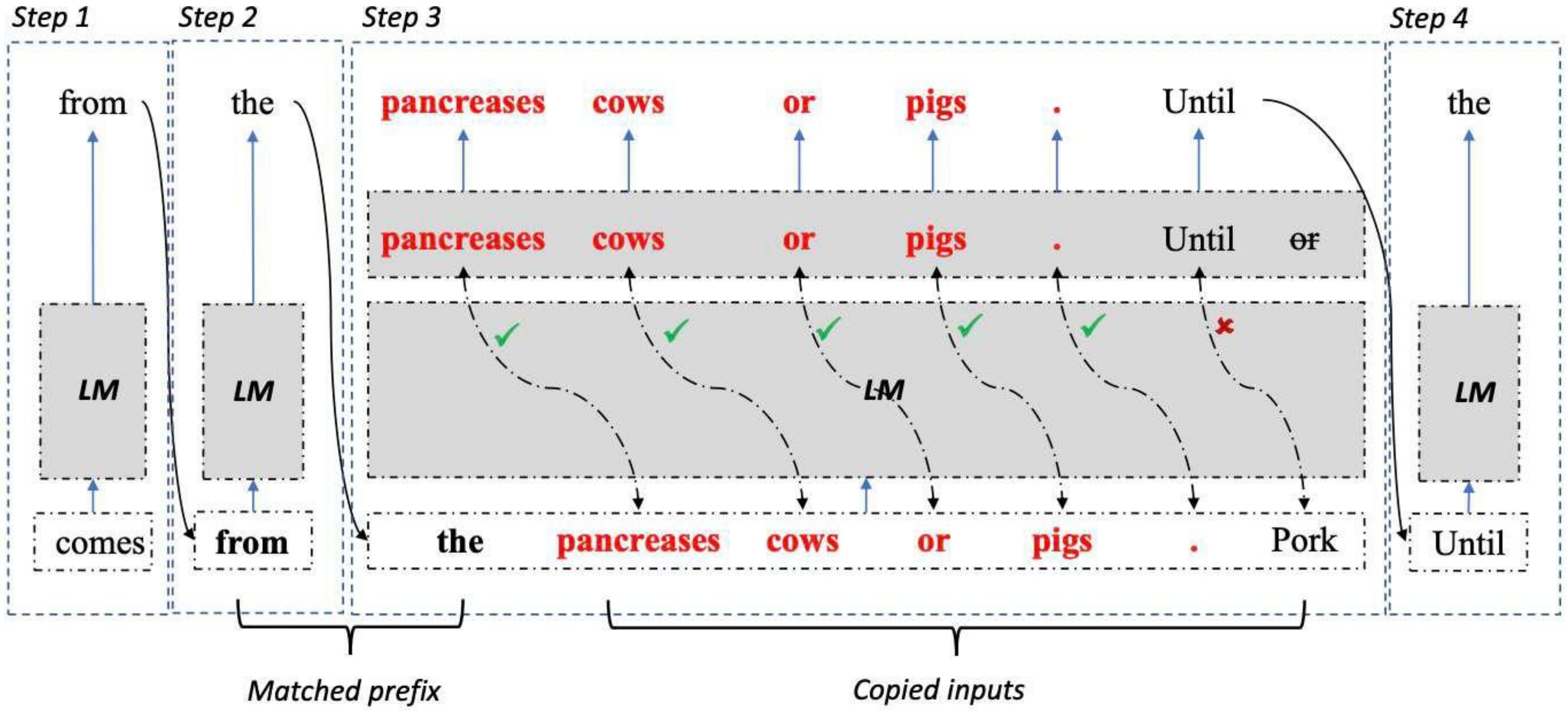
  for $t = 1 : K$ **do**
    Sample $r \sim U[0, 1]$ from a uniform distribution.
    if $r < \min\left(1, \frac{q(x|x_1,\ldots,x_{n+t-1})}{p(x|x_1,\ldots,x_{n+t-1})}\right)$, **then**
      Set $x_{n+t} \leftarrow \tilde{x}_t$ and $n \leftarrow n + 1$.
    else
      sample $x_{n+t} \sim (q(x|x_1, \ldots, x_{n+t-1}) - p(x|x_1, \ldots, x_{n+t-1}))_+$ and exit for loop.
    end if
  end for
  If all tokens $x_{n+1}, \ldots, x_{n+K}$ are accepted, sample extra token $x_{n+K+1} \sim q(x|, x_1, \ldots, x_n, x_{n+K})$ and set $n \leftarrow n + 1$.
end while

Inference with reference

Algorithm 1 LLMA Decoding.

Input: $\boldsymbol{x}$, $\boldsymbol{D} = (\boldsymbol{d}_1, \ldots, \boldsymbol{d}_n)$, n, k, N;
Output: $\boldsymbol{y}$;

$\boldsymbol{y} \leftarrow []$
while $\text{LEN}(\boldsymbol{y}) < N$ **do**
  $matched, \boldsymbol{d}, pos \leftarrow \text{MATCH\_NGRAMS}(\boldsymbol{y}, \boldsymbol{D}, n)$
  if $\neg matched$ **then**
    $o = \text{LLM}(\boldsymbol{x}, \boldsymbol{y})$
    $\text{APPEND}(\boldsymbol{y}, o)$
    continue
  end if
  $(o_0, o_1, \ldots, o_k) \leftarrow \text{LLM}(\boldsymbol{x}, \boldsymbol{y}, d_{pos}, \ldots, d_{pos+k-1})$
  $\text{APPEND}(\boldsymbol{y}, o_0)$
  for i **in** $0, \ldots, k-1$ **do**
    if o_i **neq** d_{pos+i} **then**
      break
    end if
    $\text{APPEND}(\boldsymbol{y}, o_{i+1})$
  end for
end while



Why speculative sampling is bad

- Optimize only MBU
- Work only with batch_size=1
- Don't optimize memory(otherwise)
- hard to implement

Conclusion

- Don’t forget about MBU vs MFU!!!
- Smoothquant is universal, gptq/aqlm if you have small rps/need to host huge models on 1 GPU
- If you have reword model use it!, otherwise hard-label KD if you have a lot of compute, slim if not
- Don’t forget about architecture optimizing